

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

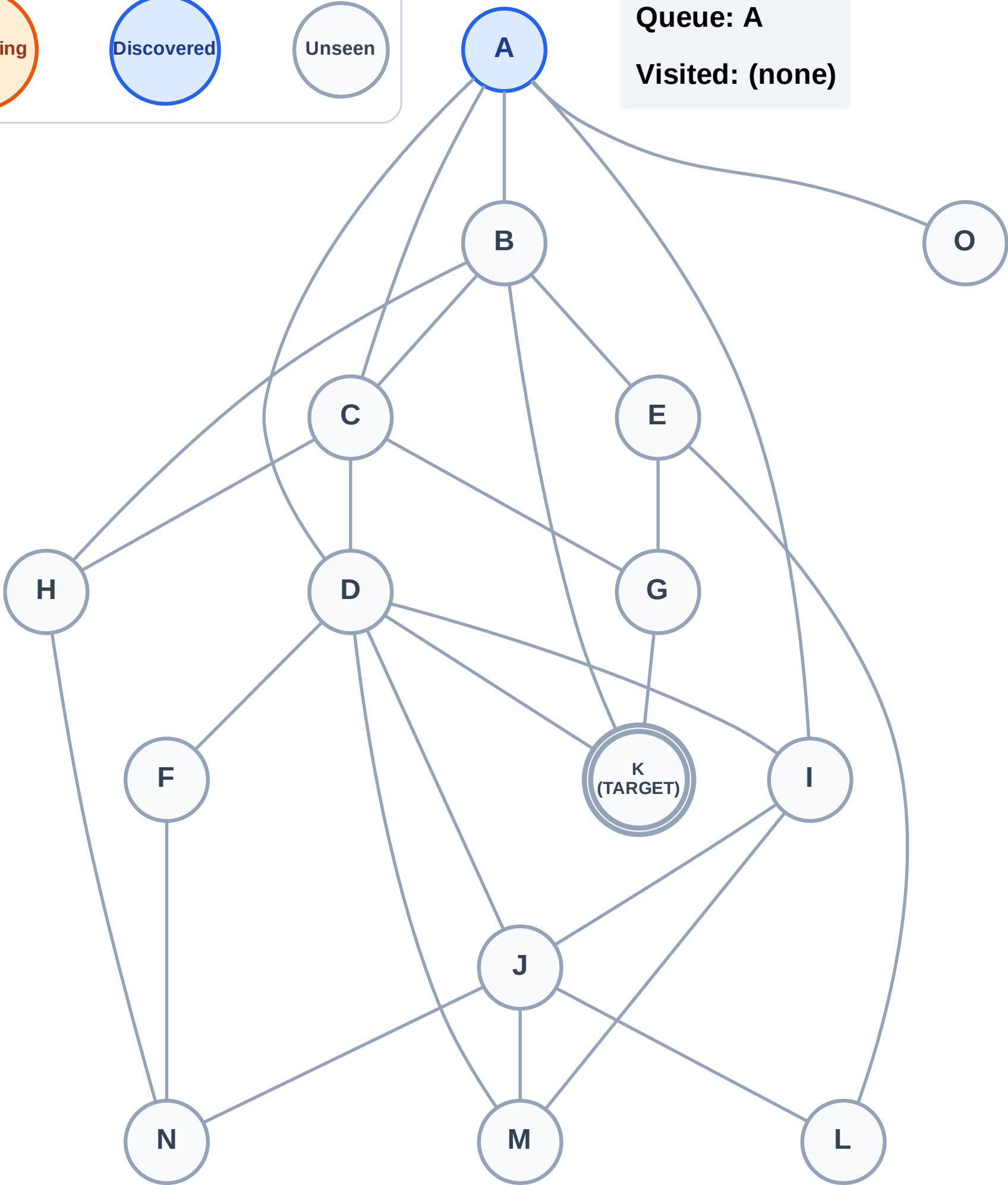
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

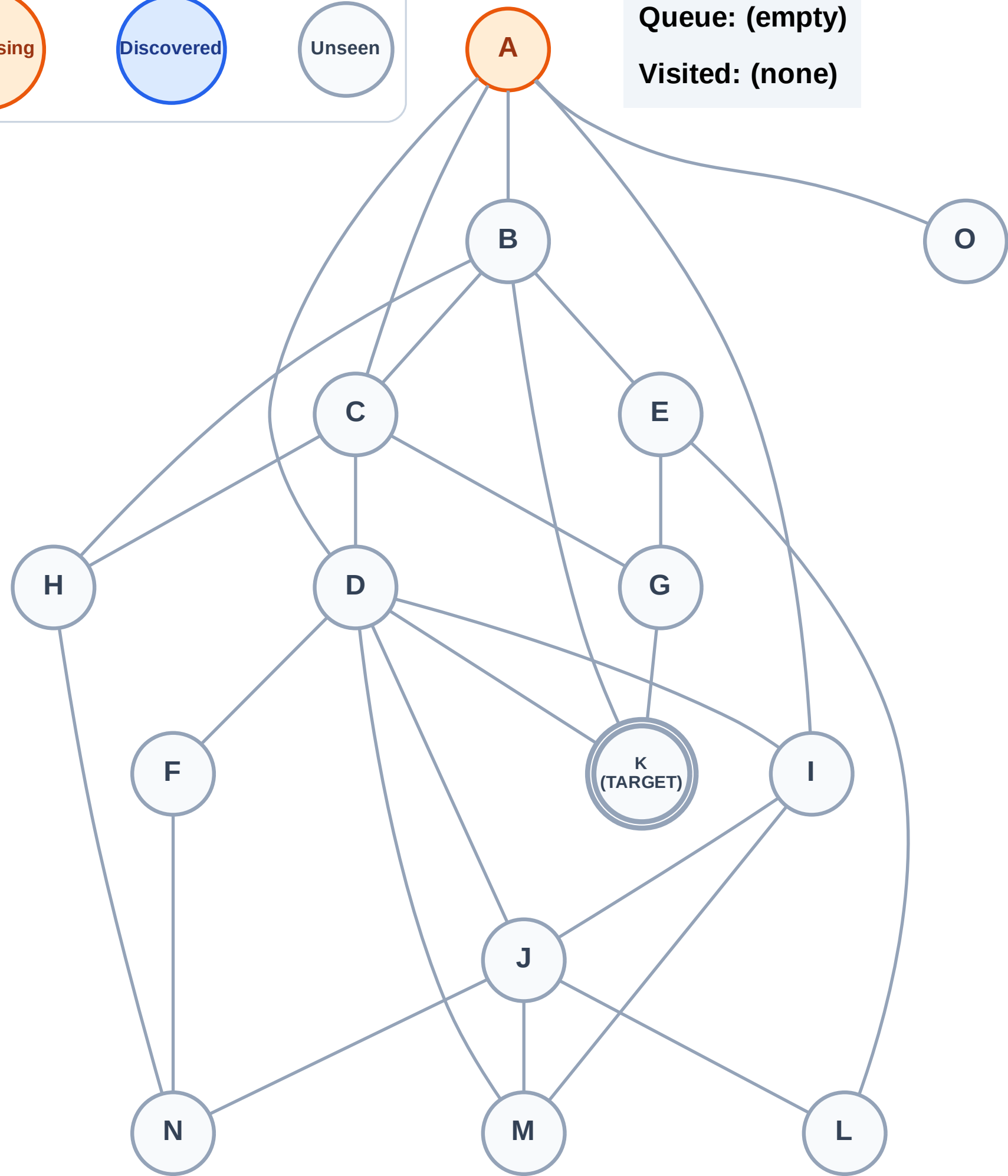
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

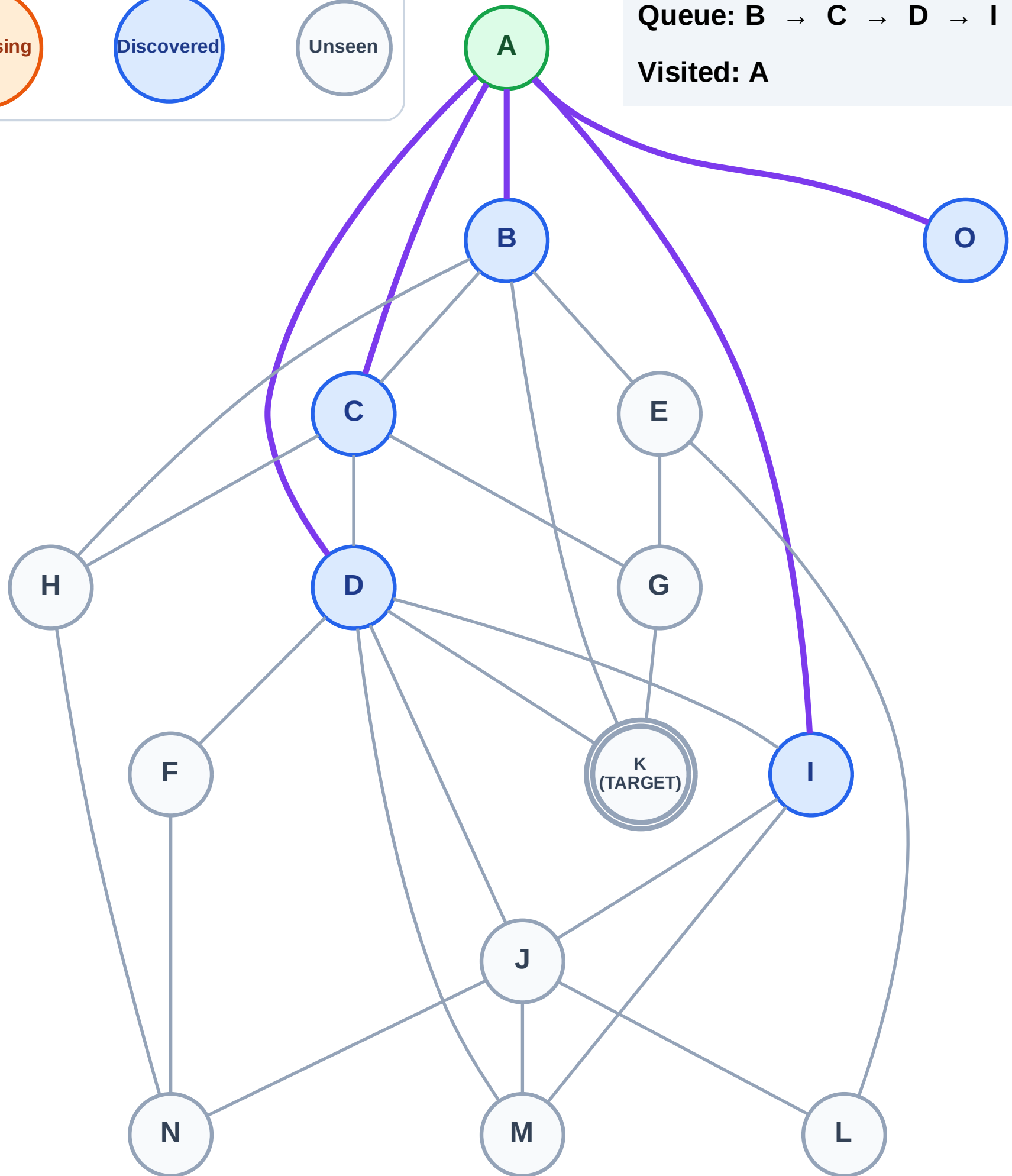
Step 3: Discover B, C, D, I, O from A

Legend



Queue: B → C → D → I → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

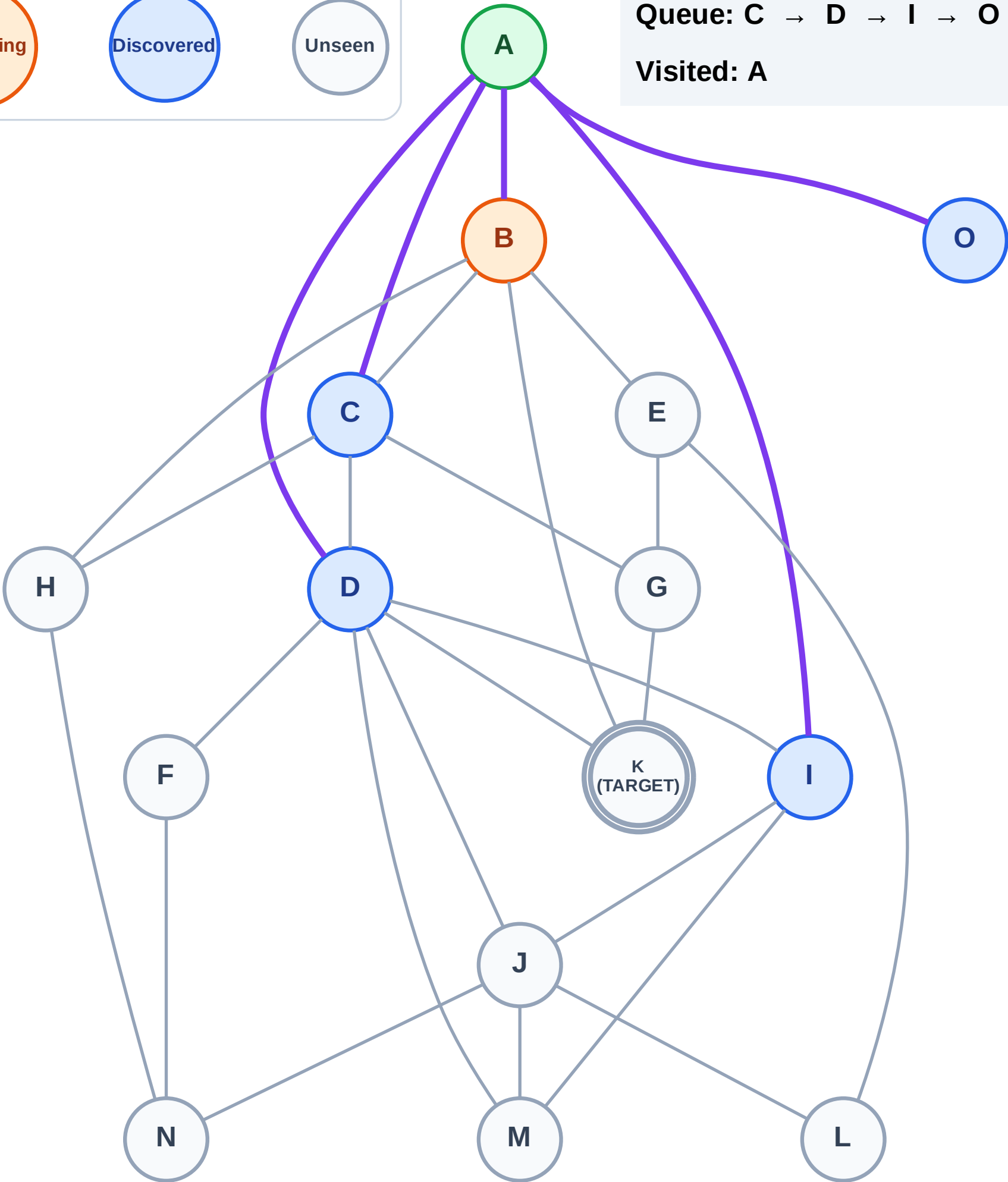
Processing

Discovered

Unseen

Queue: C → D → I → O

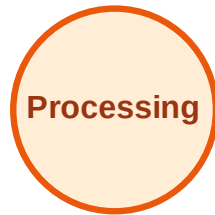
Visited: A



BFS Traversal

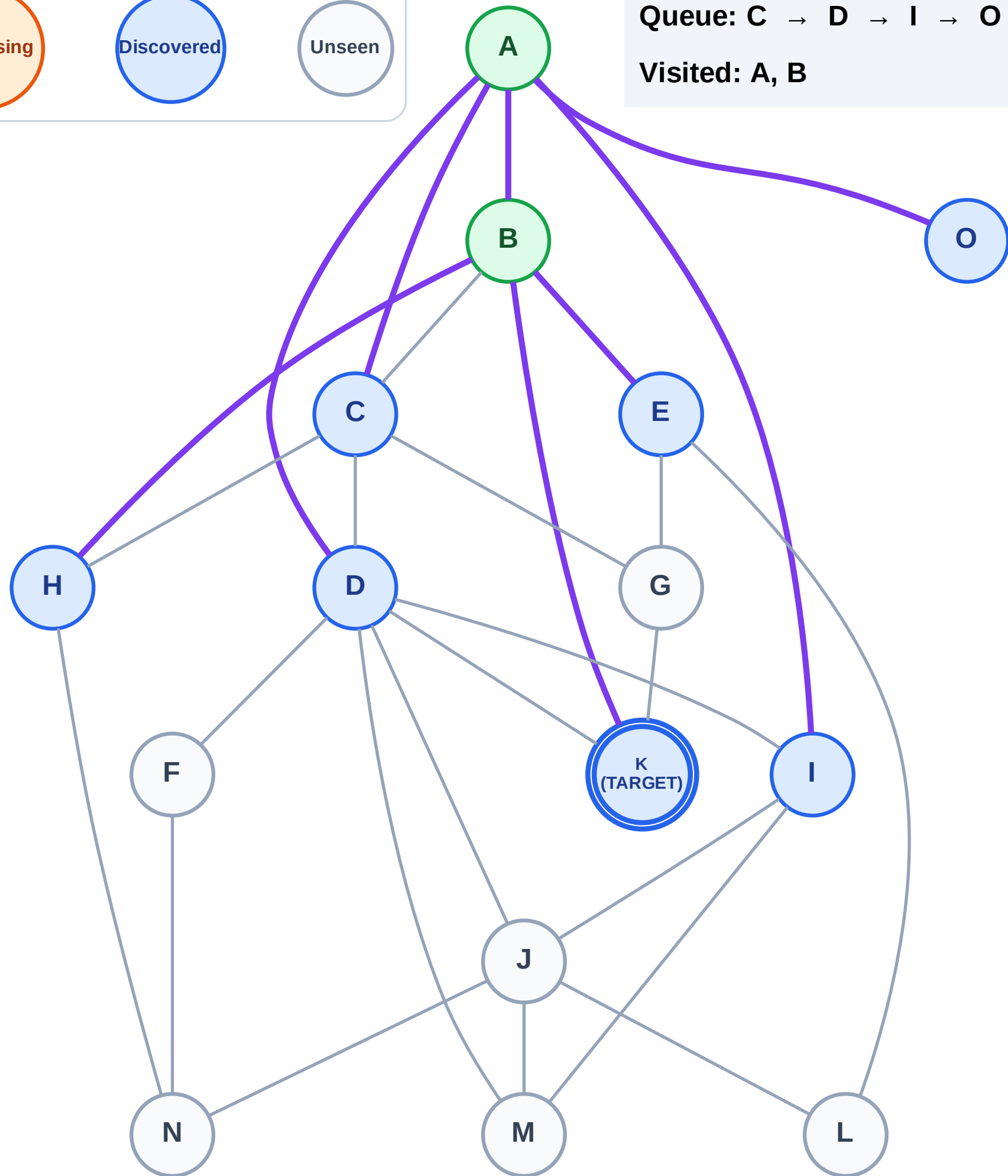
Step 5: Discover E, H, K from B

Legend



Queue: C → D → I → O → E → H → K

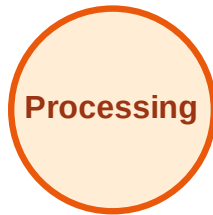
Visited: A, B



BFS Traversal

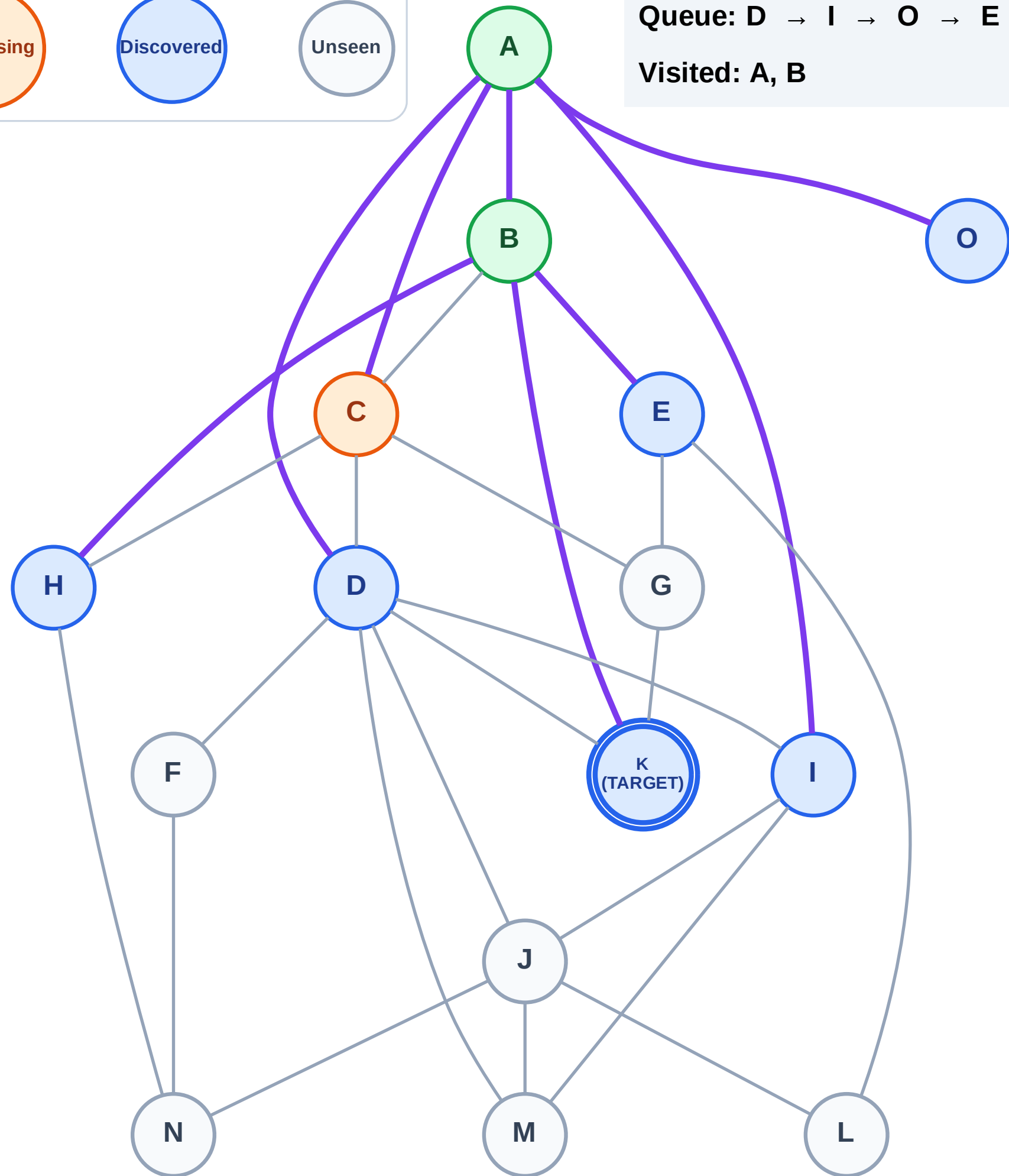
Step 6: Process node C

Legend

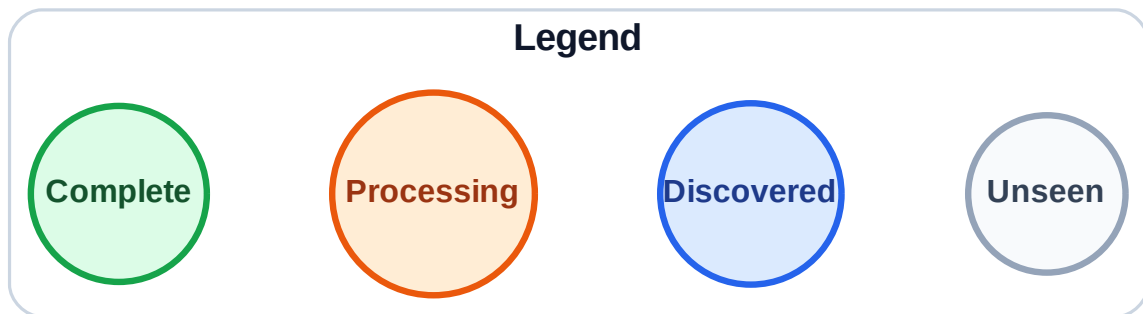


Queue: D → I → O → E → H → K

Visited: A, B

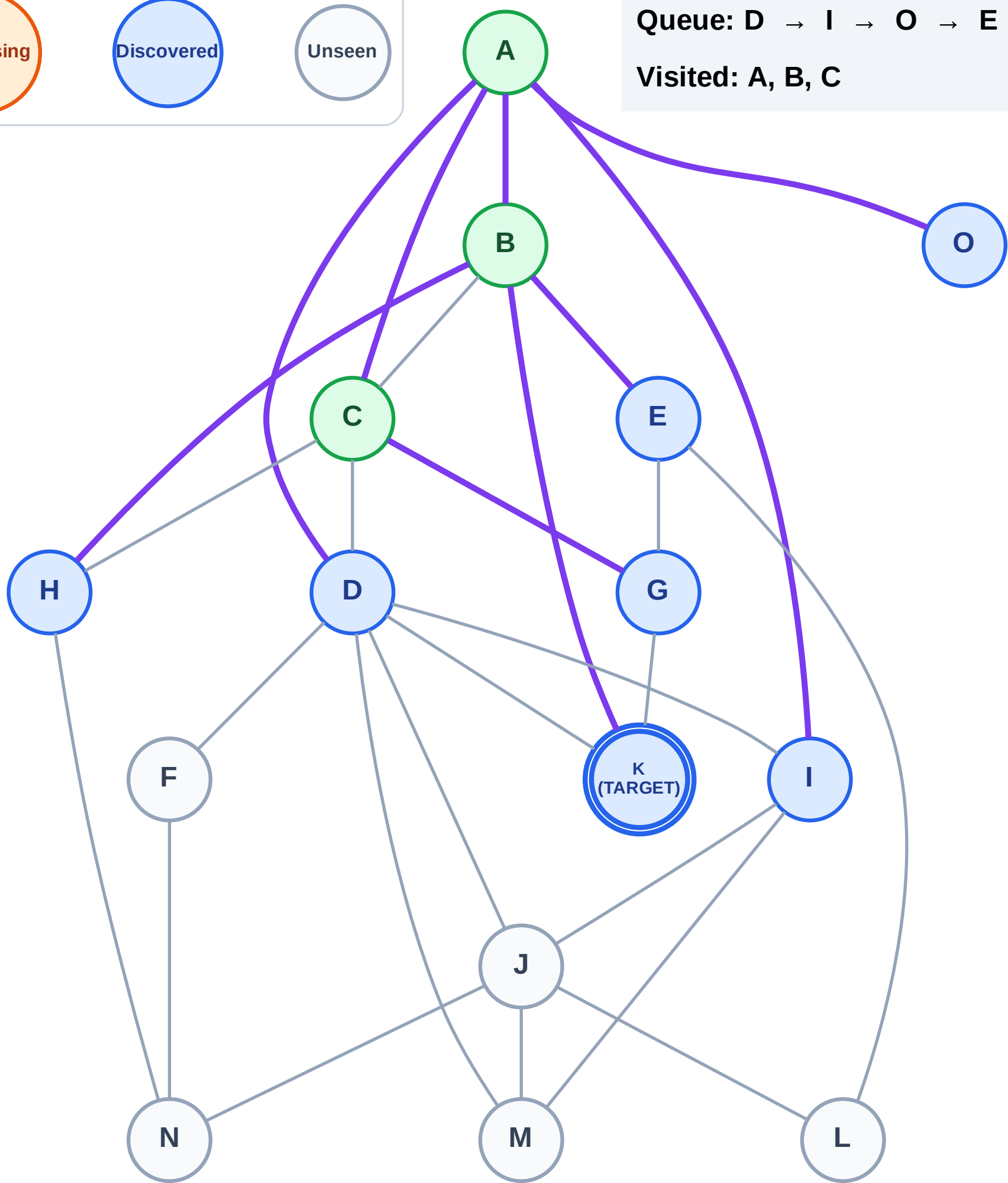


Step 7: Discover G from C



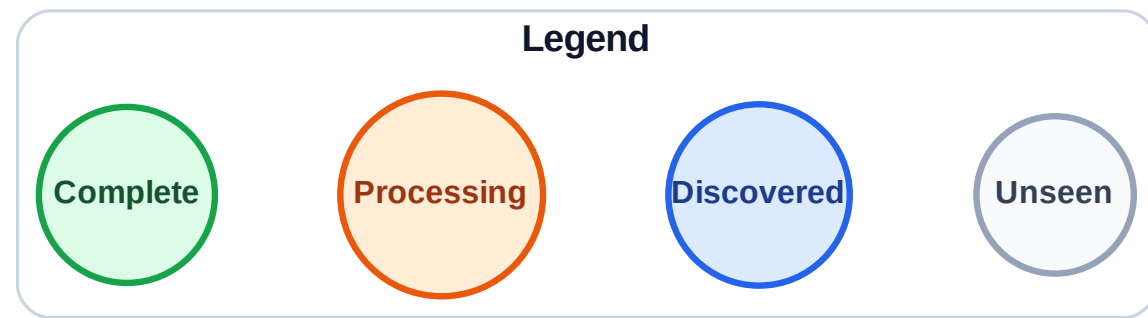
Queue: D → I → O → E → H → K → G

Visited: A, B, C



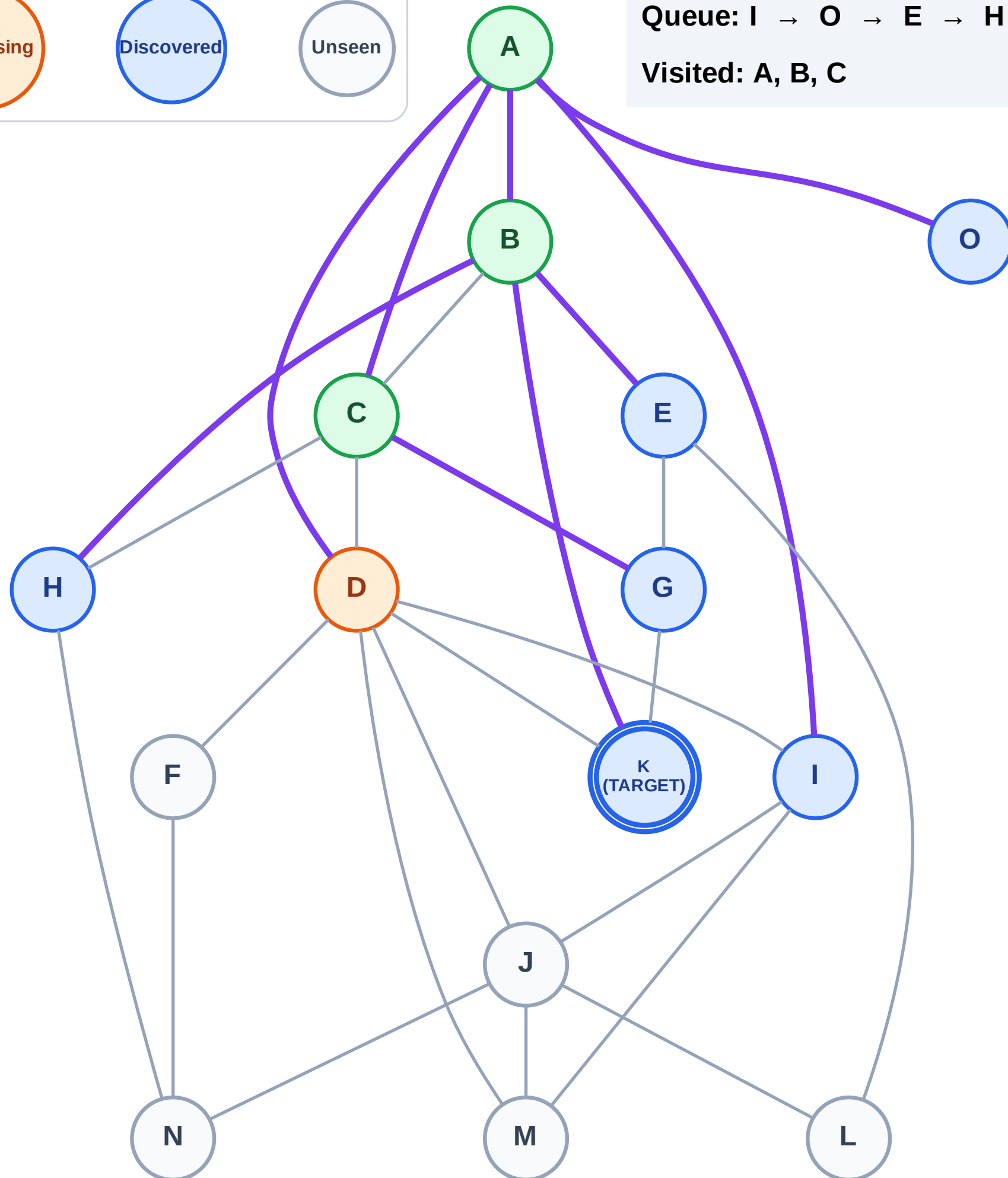
BFS Traversal

Step 8: Process node D



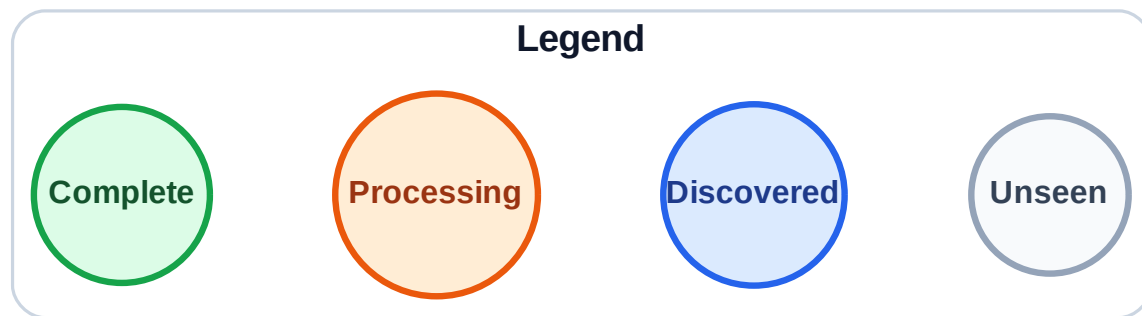
Queue: I → O → E → H → K → G

Visited: A, B, C



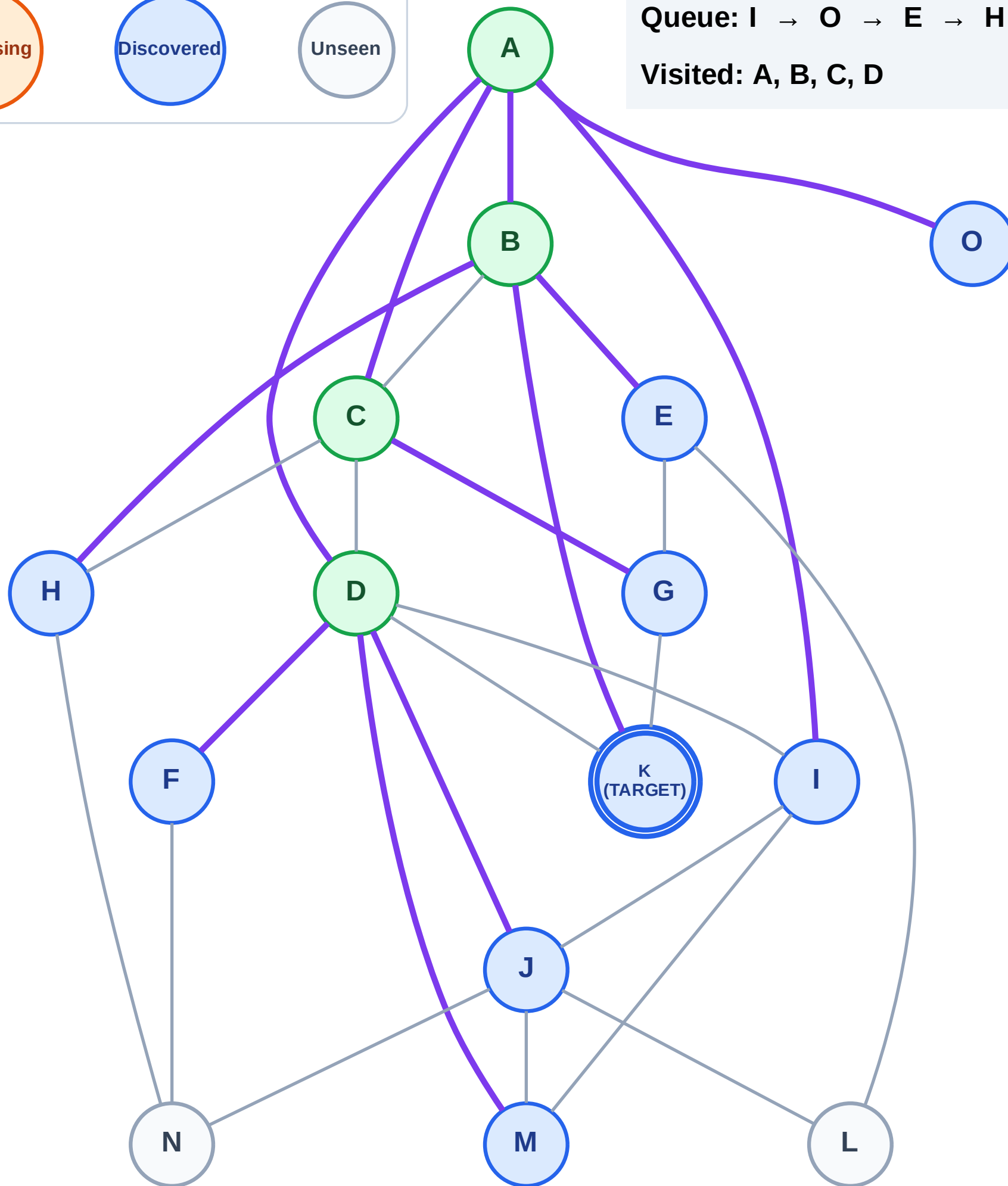
BFS Traversal

Step 9: Discover F, J, M from D



Queue: I → O → E → H → K → G → F → J → M

Visited: A, B, C, D



BFS Traversal

Step 10: Process node I

Legend

Complete

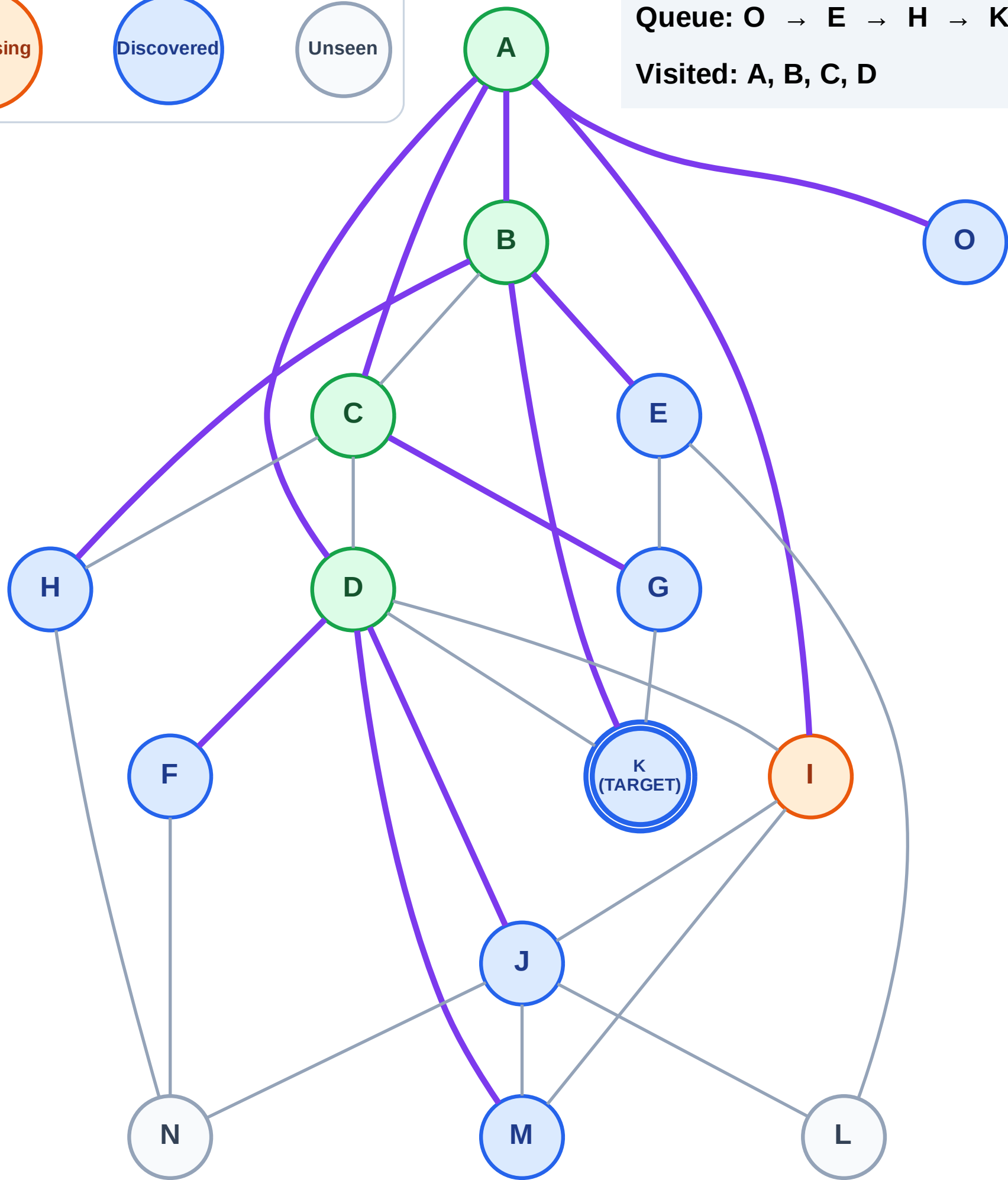
Processing

Discovered

Unseen

Queue: O → E → H → K → G → F → J → M

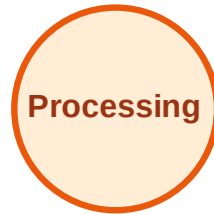
Visited: A, B, C, D



BFS Traversal

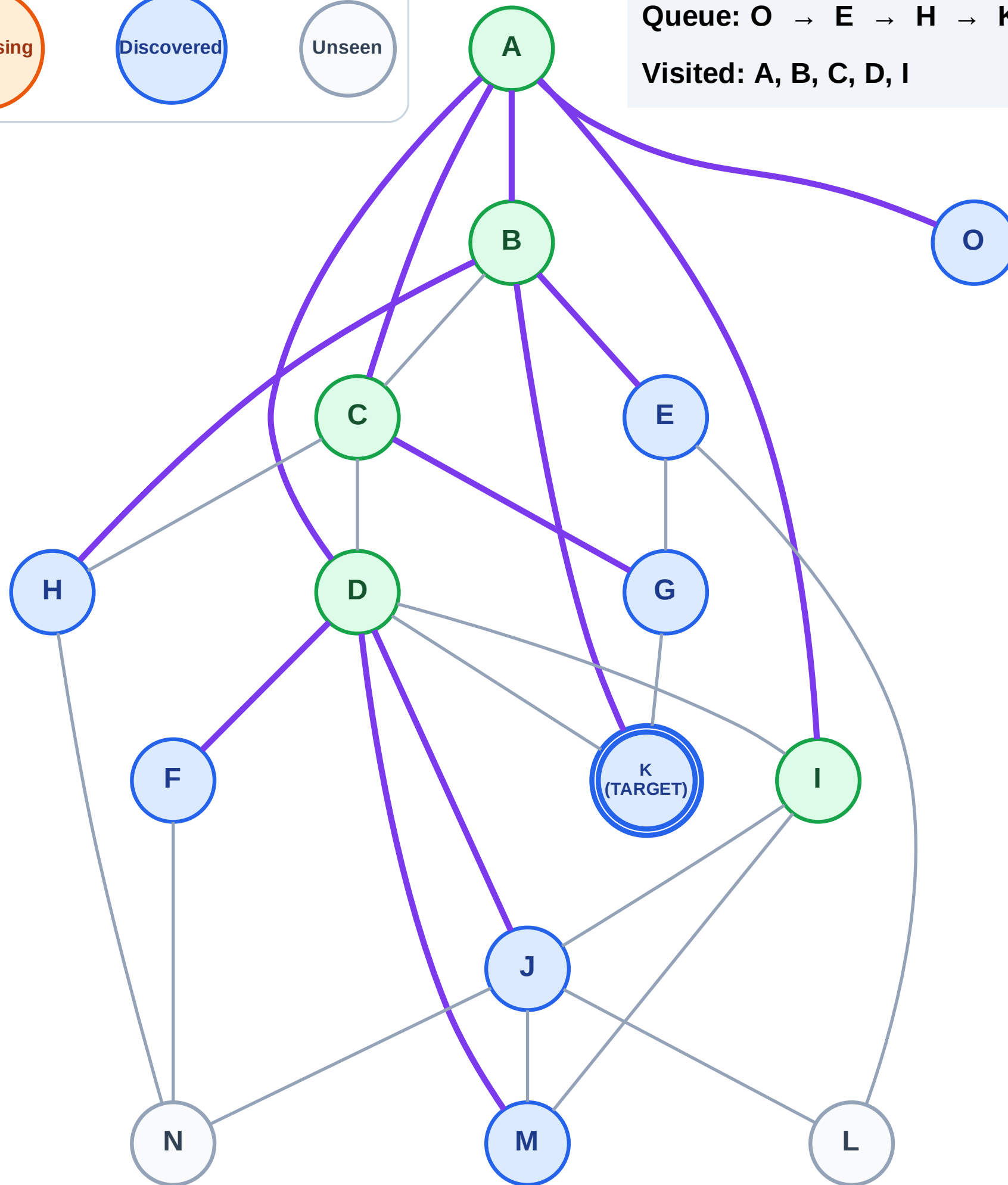
Step 11: No new neighbors from I

Legend



Queue: O → E → H → K → G → F → J → M

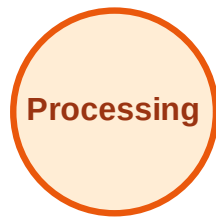
Visited: A, B, C, D, I



BFS Traversal

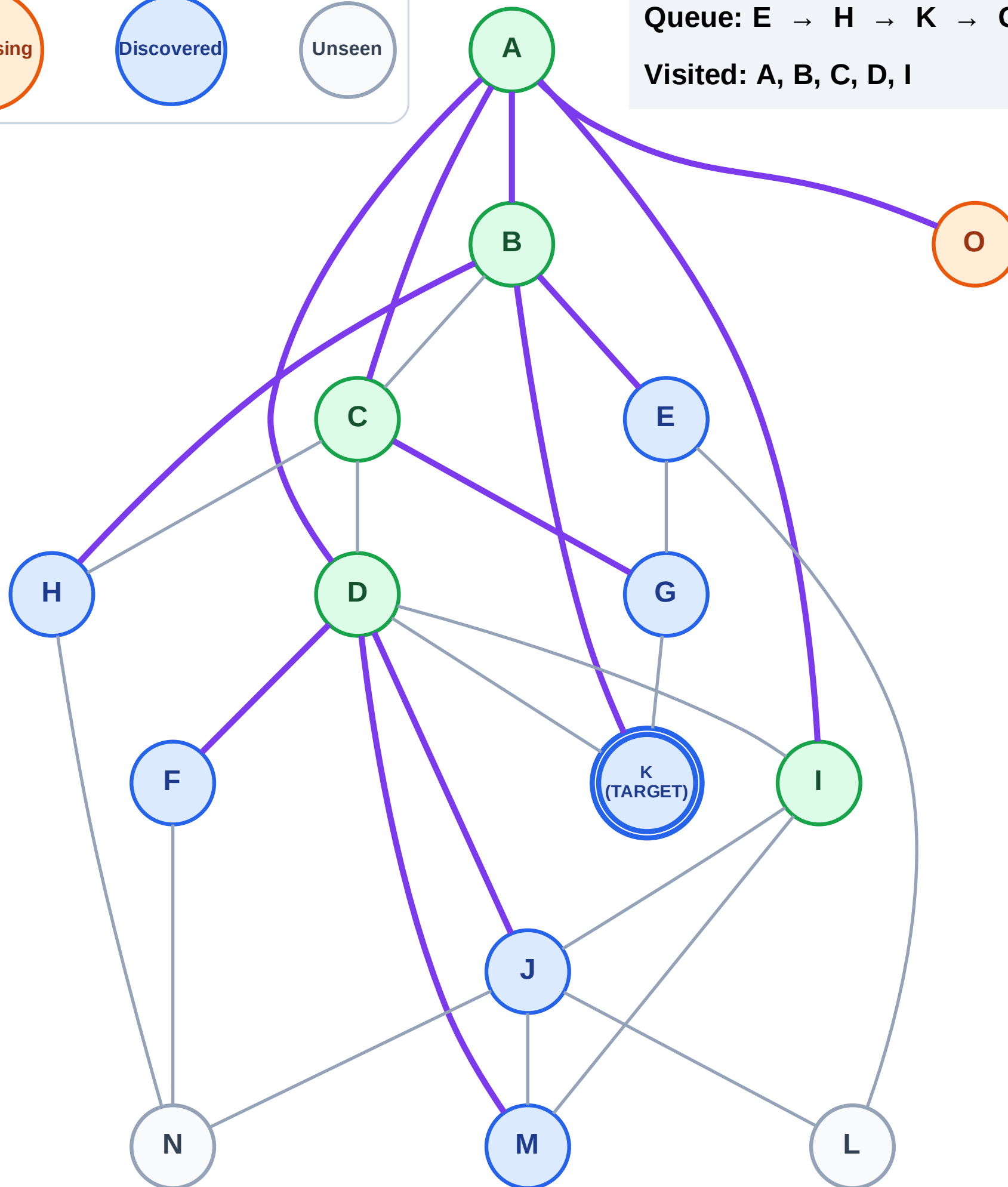
Step 12: Process node O

Legend



Queue: E → H → K → G → F → J → M

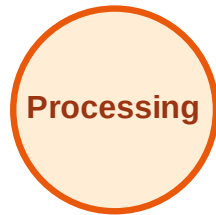
Visited: A, B, C, D, I



BFS Traversal

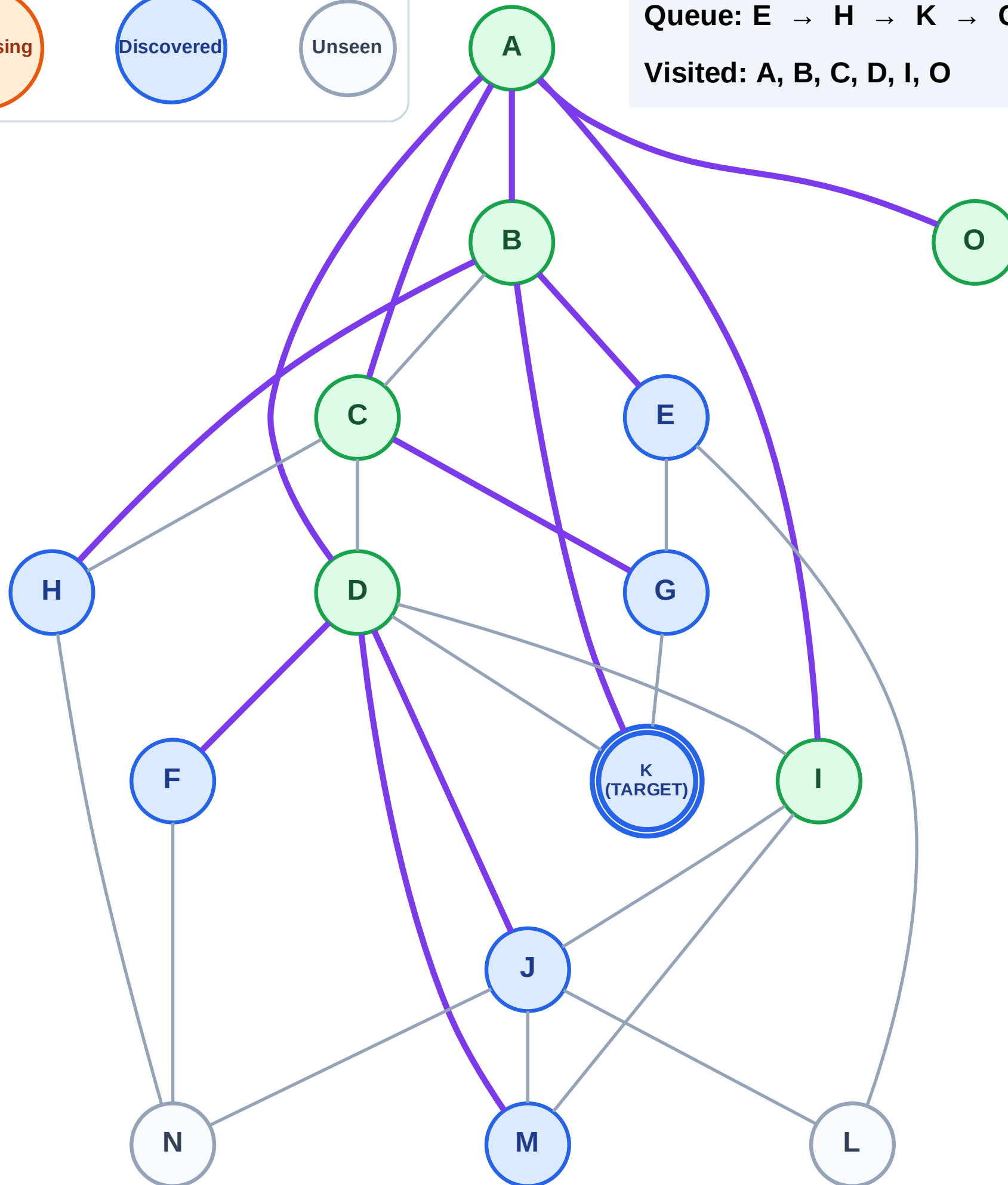
Step 13: No new neighbors from O

Legend



Queue: E → H → K → G → F → J → M

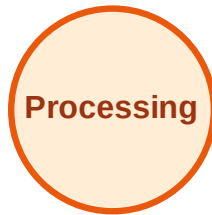
Visited: A, B, C, D, I, O



BFS Traversal

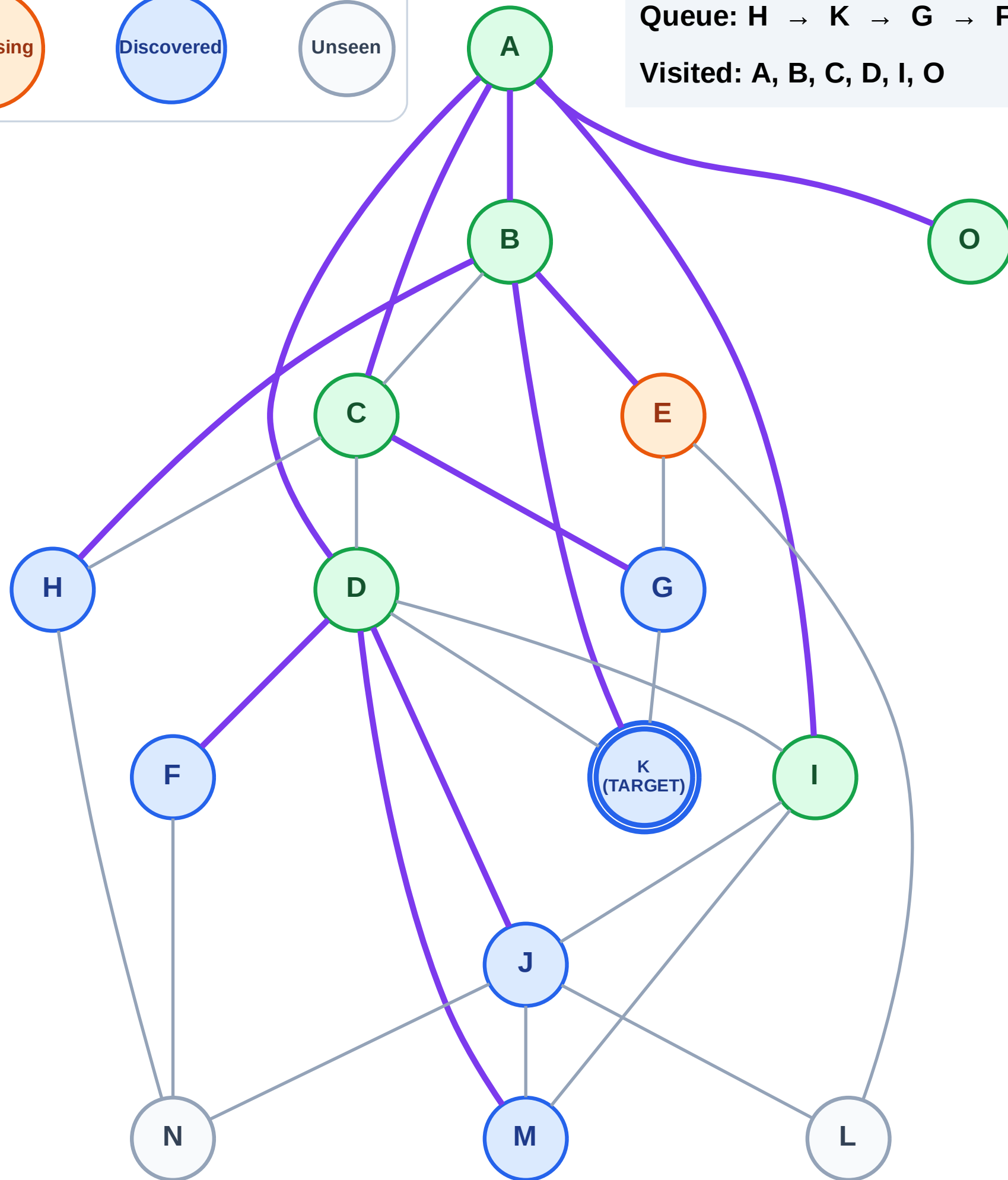
Step 14: Process node E

Legend



Queue: H → K → G → F → J → M

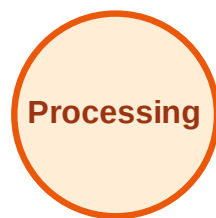
Visited: A, B, C, D, I, O



BFS Traversal

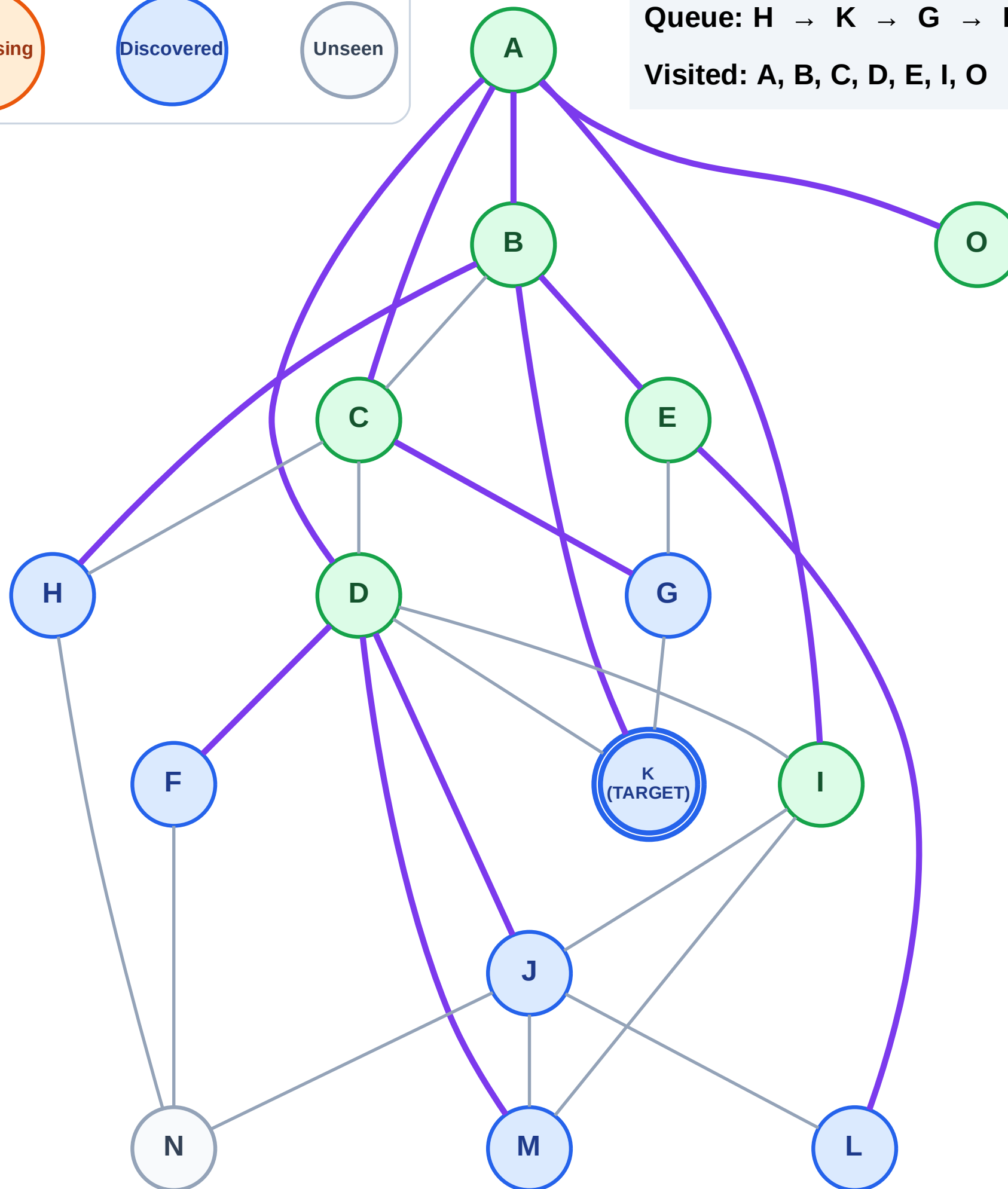
Step 15: Discover L from E

Legend



Queue: H → K → G → F → J → M → L

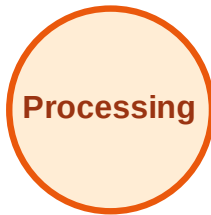
Visited: A, B, C, D, E, I, O



BFS Traversal

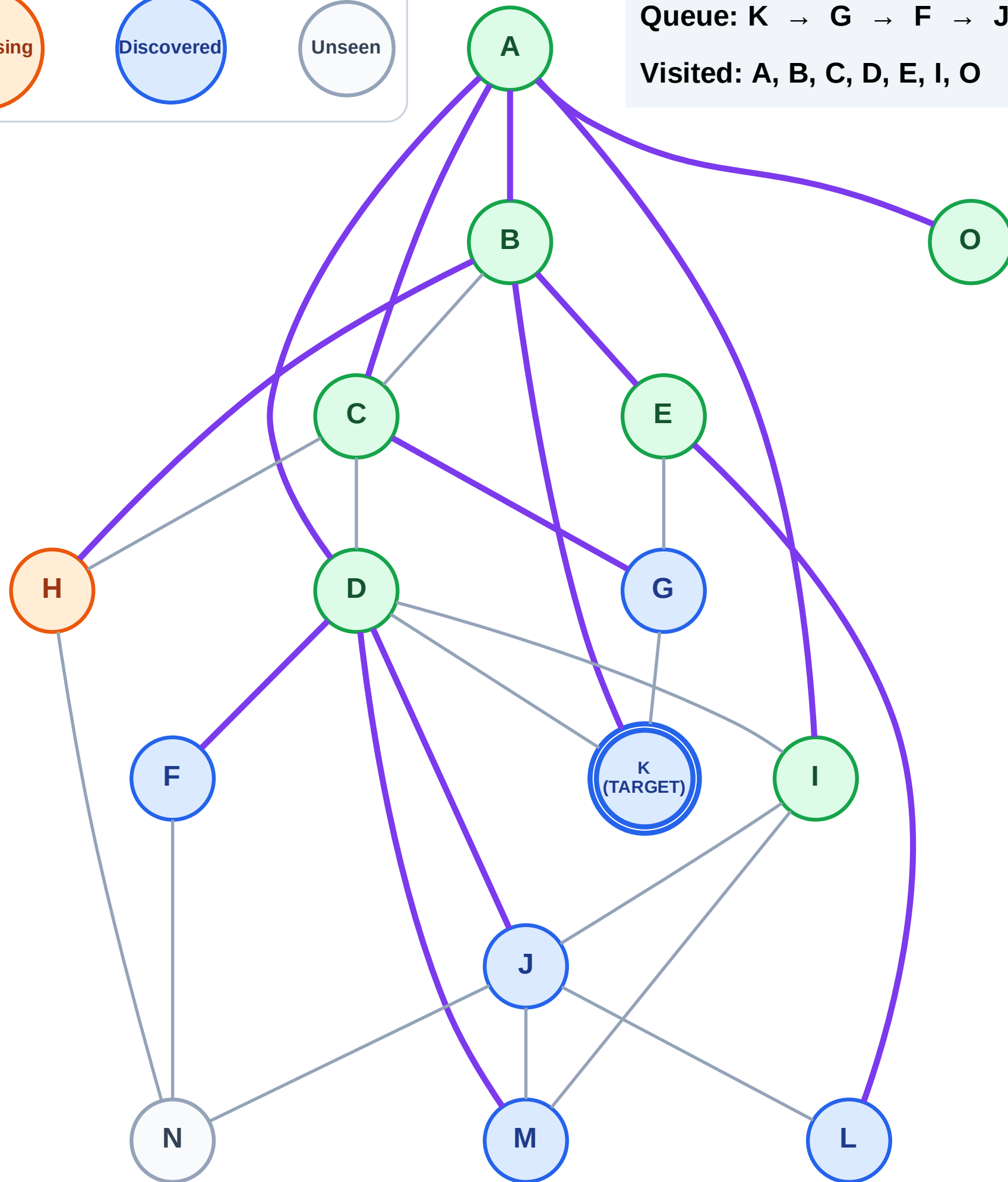
Step 16: Process node H

Legend



Queue: K → G → F → J → M → L

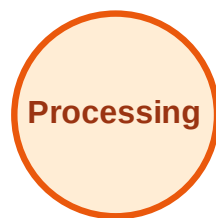
Visited: A, B, C, D, E, I, O



BFS Traversal

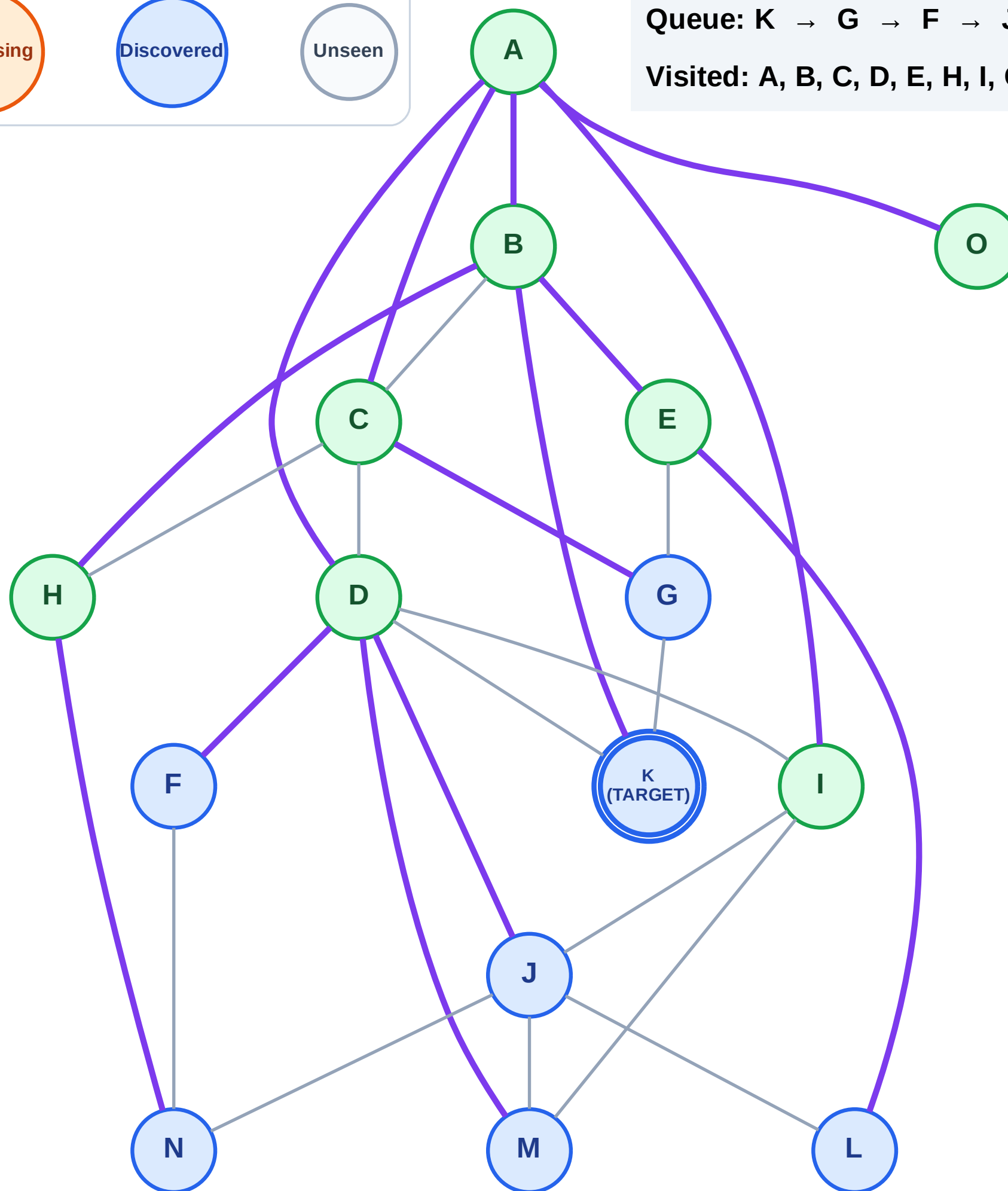
Step 17: Discover N from H

Legend



Queue: K → G → F → J → M → L → N

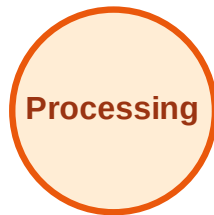
Visited: A, B, C, D, E, H, I, O



BFS Traversal

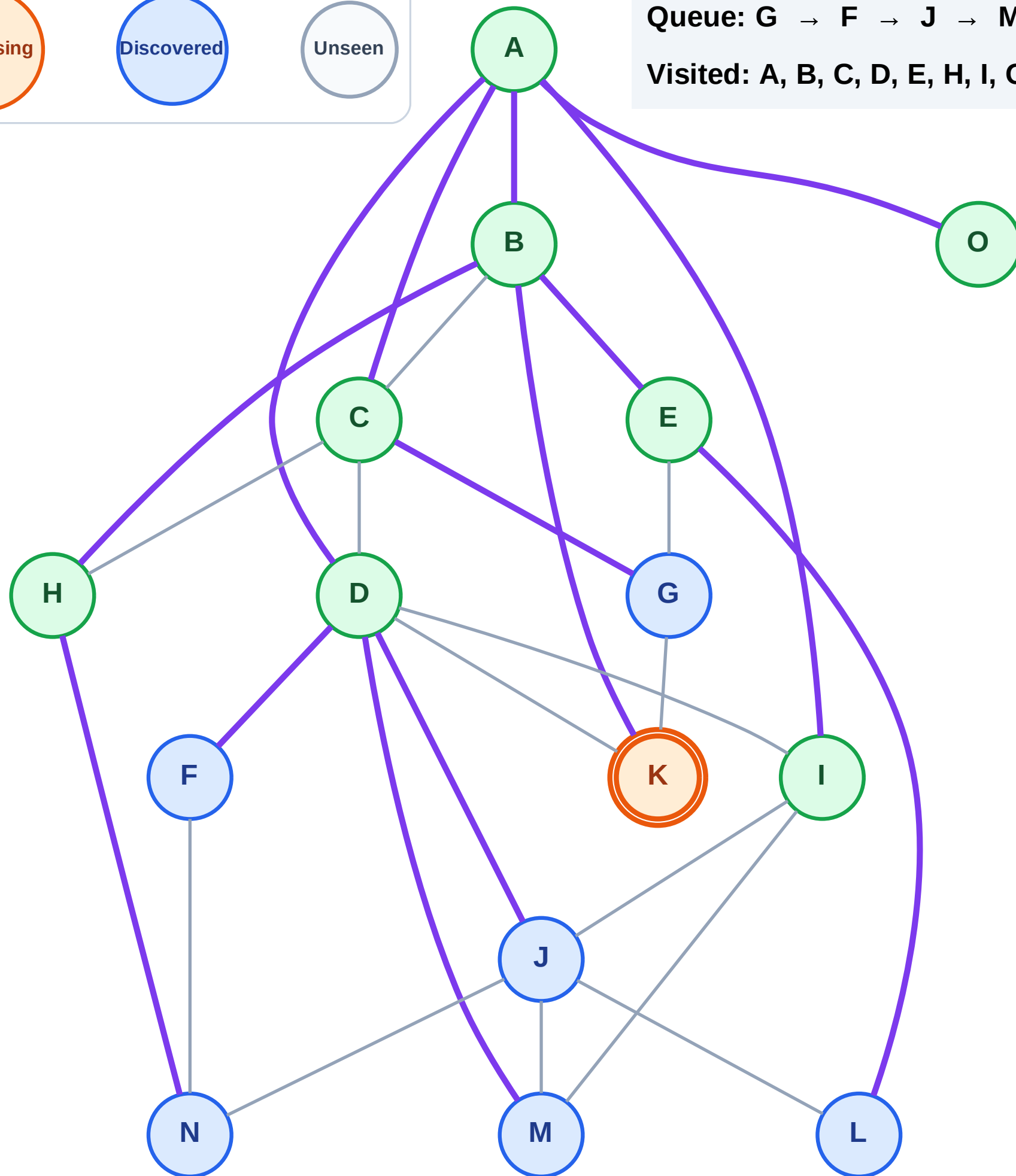
Step 18: Process node K

Legend



Queue: G → F → J → M → L → N

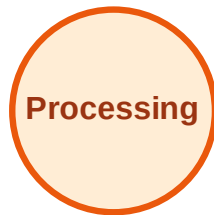
Visited: A, B, C, D, E, H, I, O



BFS Traversal

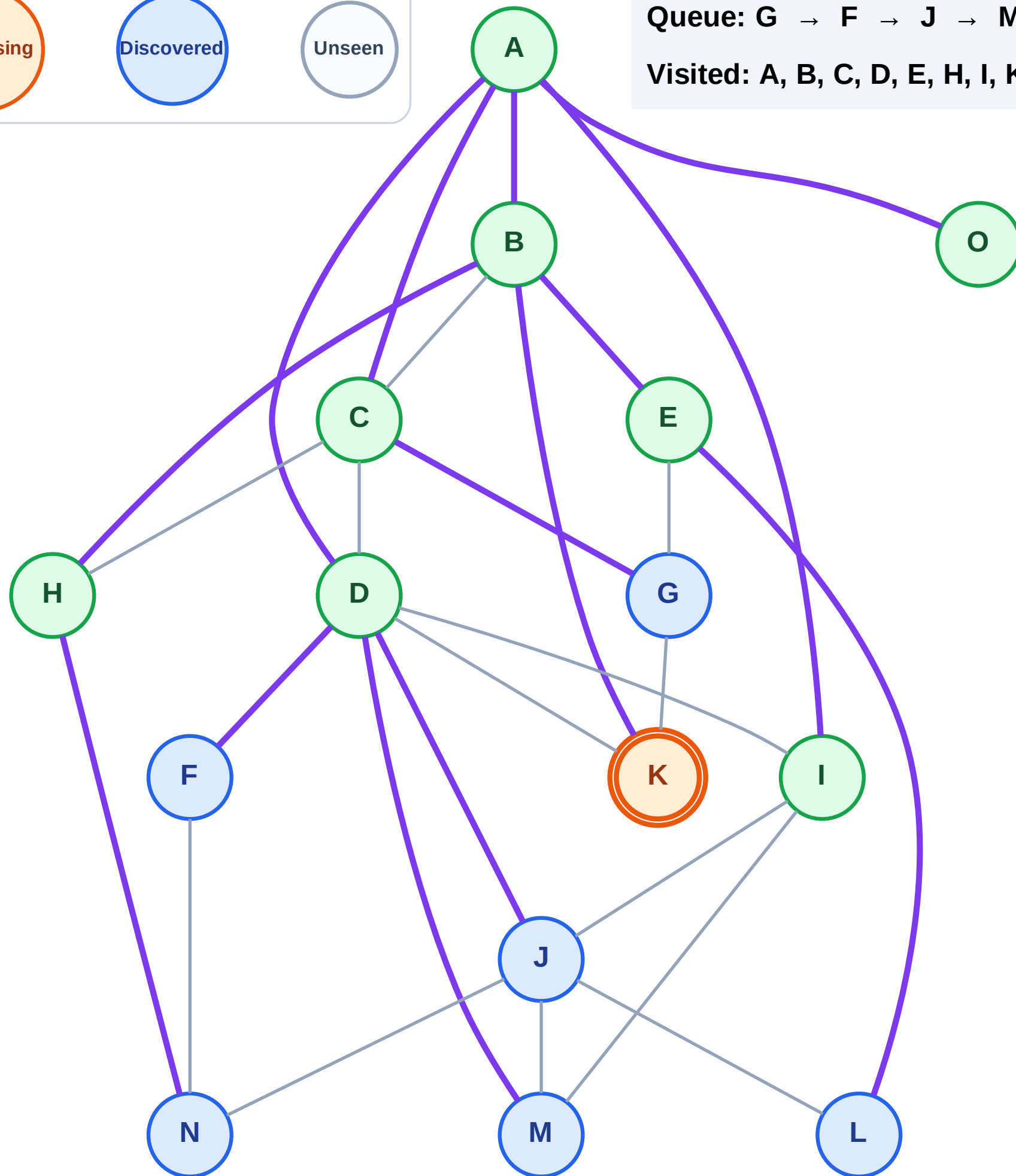
Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: G → F → J → M → L → N

Visited: A, B, C, D, E, H, I, K, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

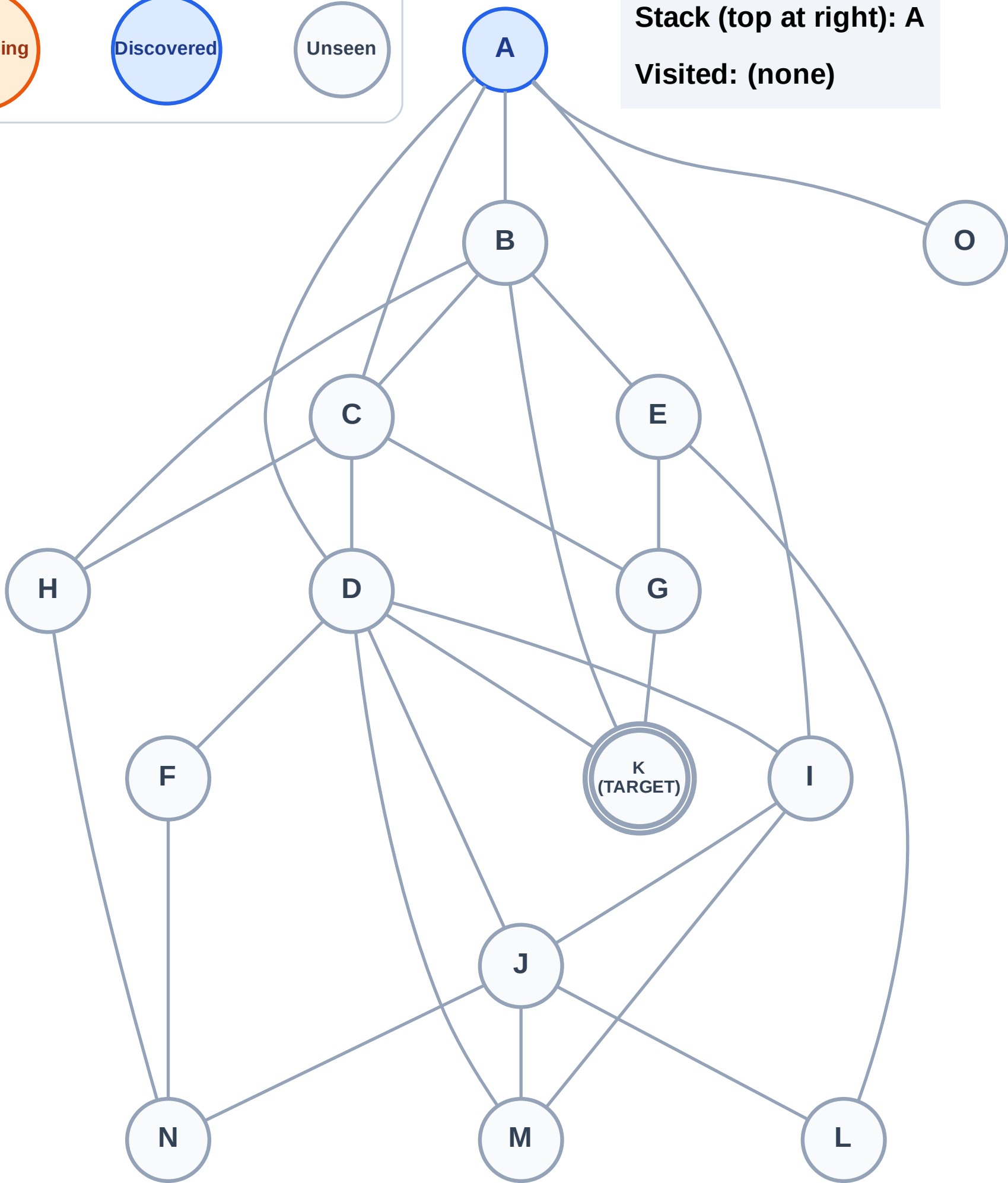
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

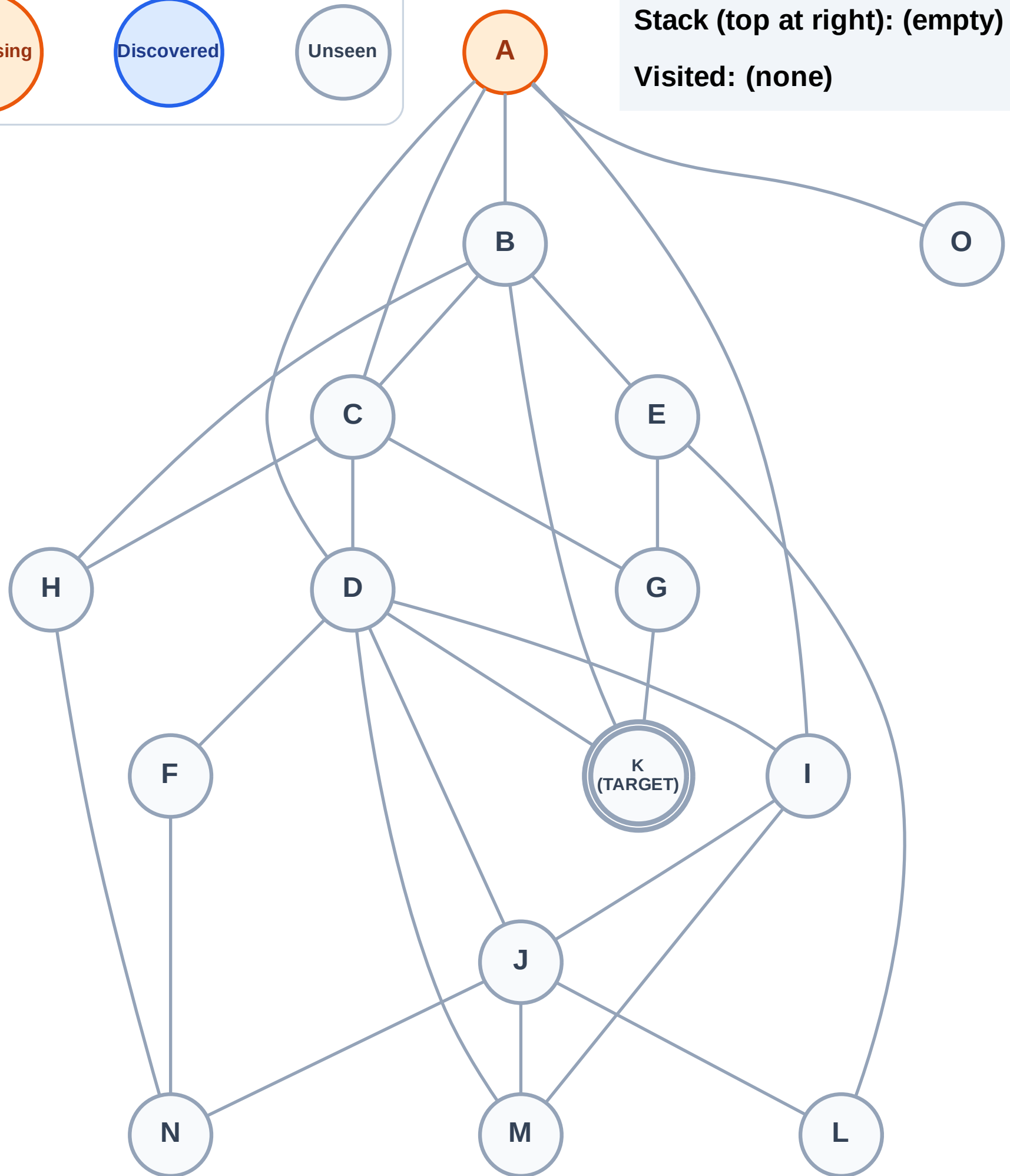
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover O, I, D, C, B from A

Legend

Complete

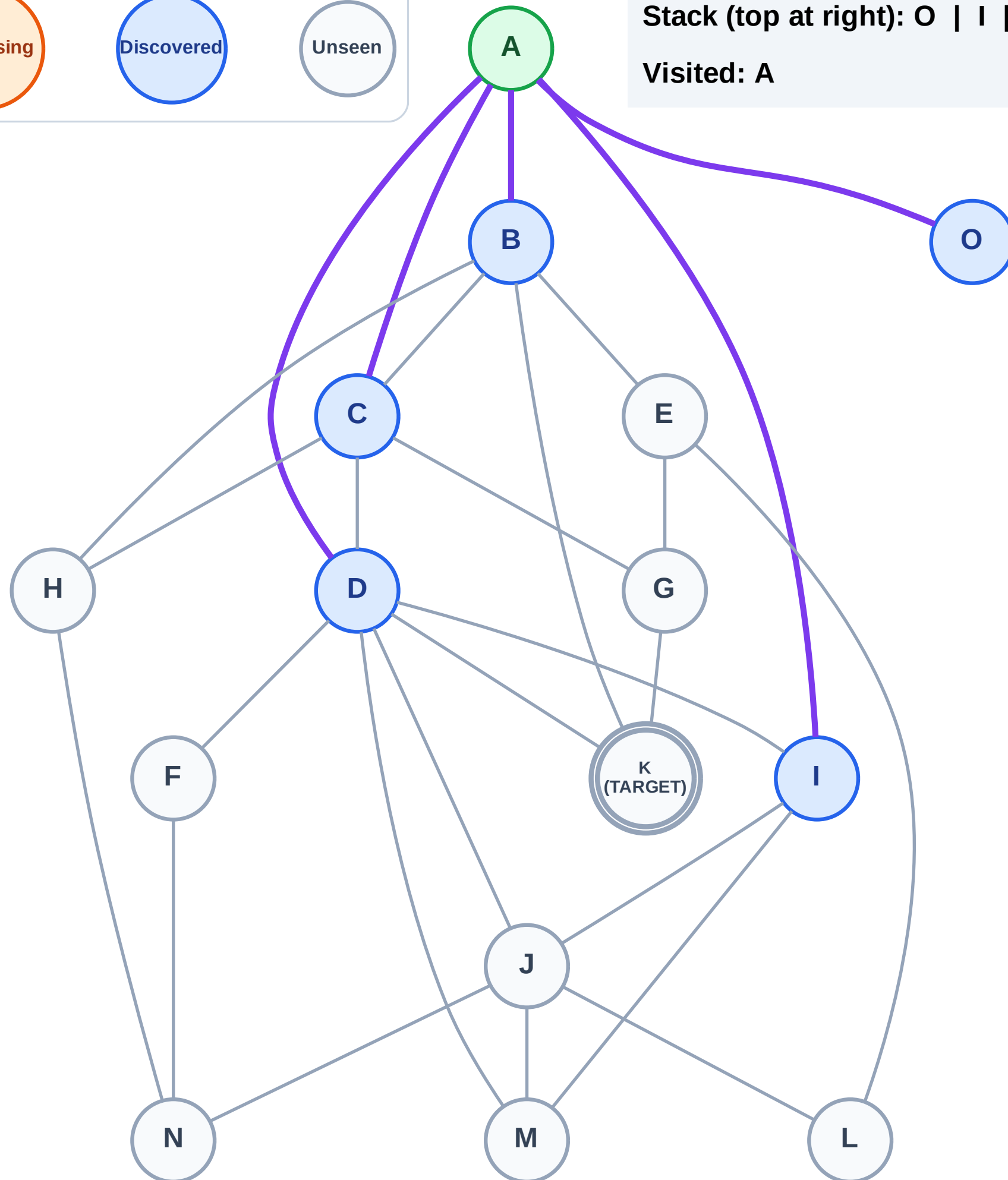
Processing

Discovered

Unseen

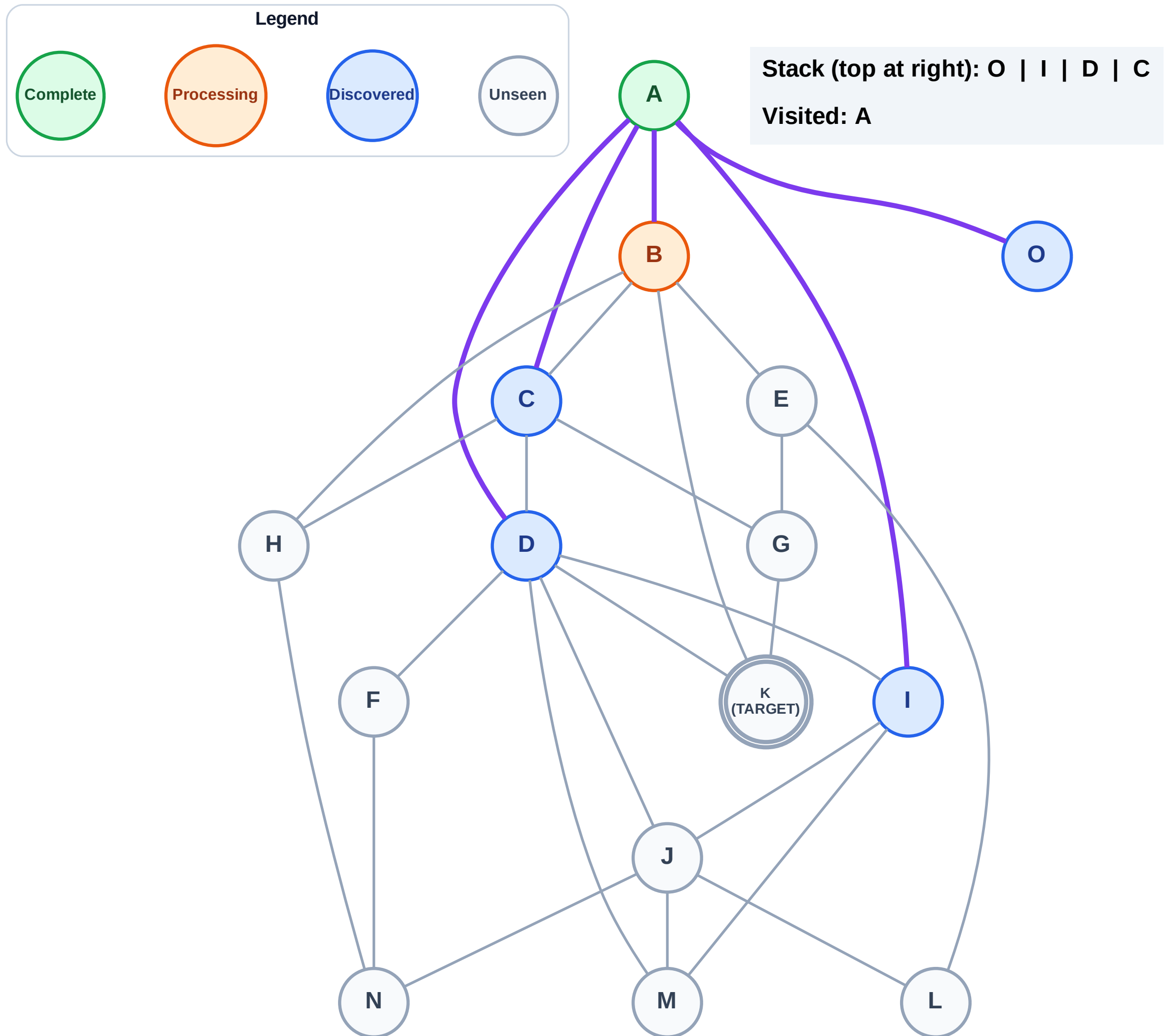
Stack (top at right): O | I | D | C | B

Visited: A



Step 4: Process node B

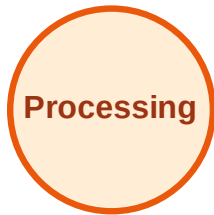
Step 4: Process node B



DFS Traversal

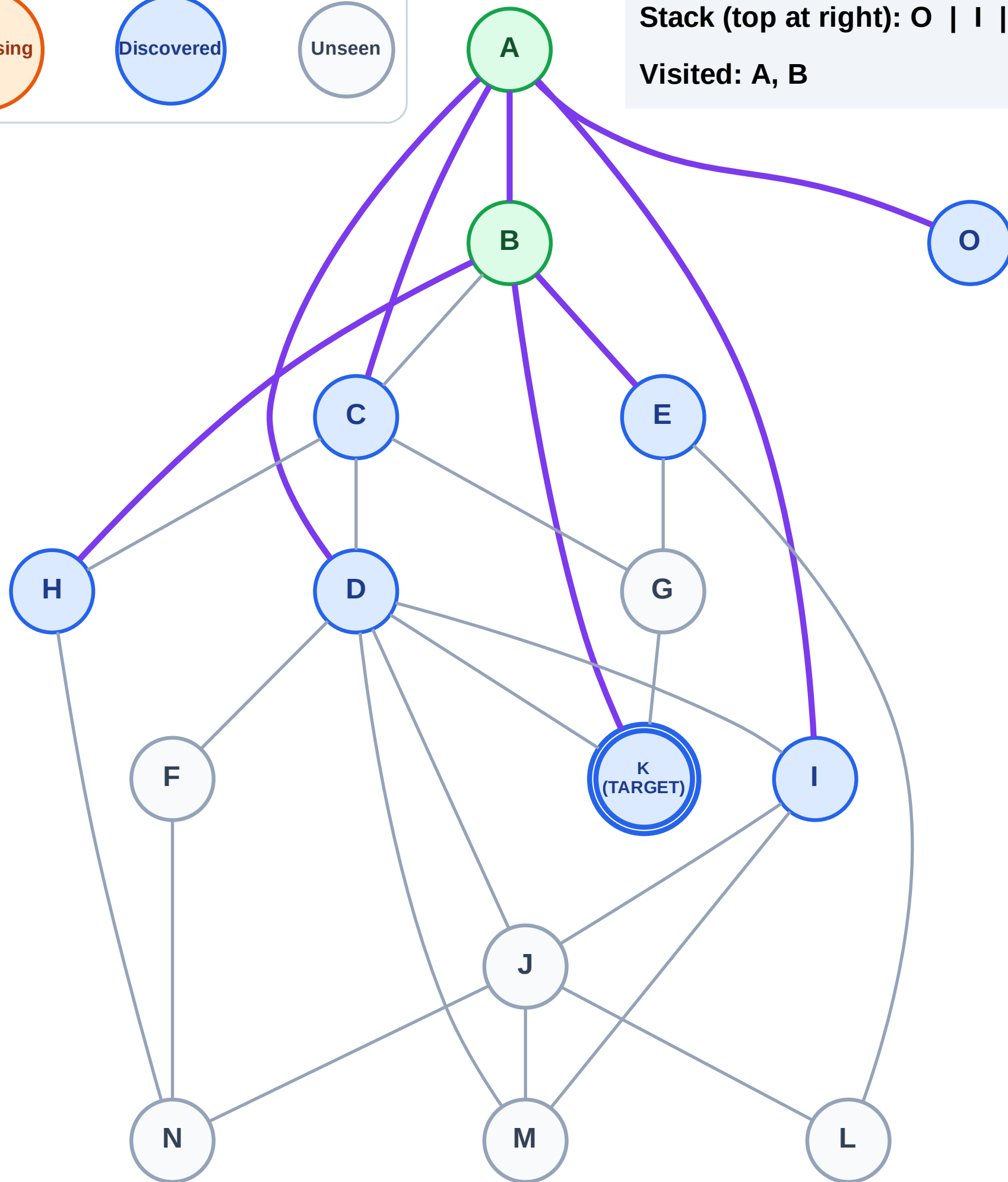
Step 5: Discover K, H, E from B

Legend



Stack (top at right): O | I | D | C | K | H | E

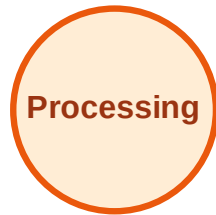
Visited: A, B



DFS Traversal

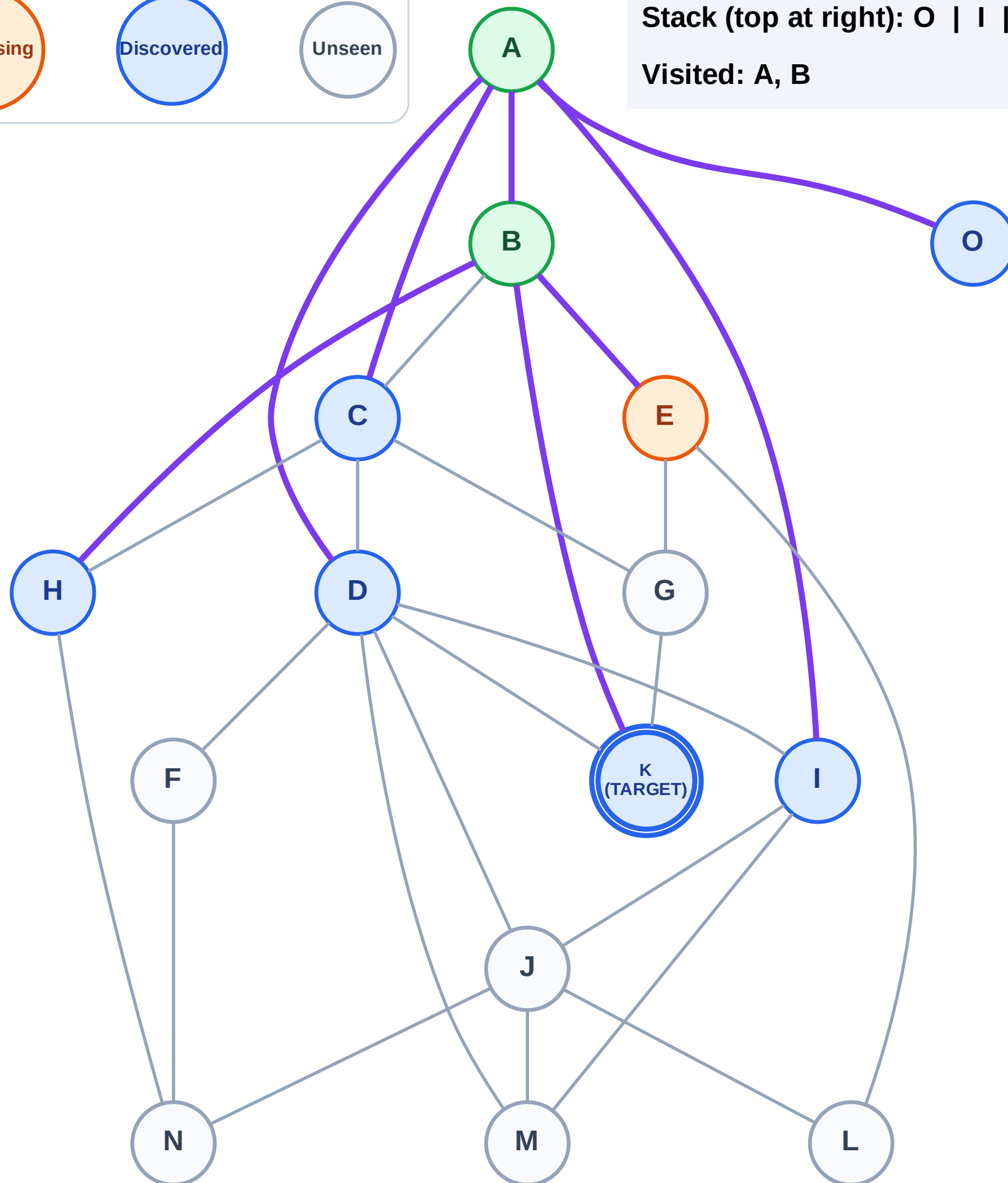
Step 6: Process node E

Legend



Stack (top at right): O | I | D | C | K | H

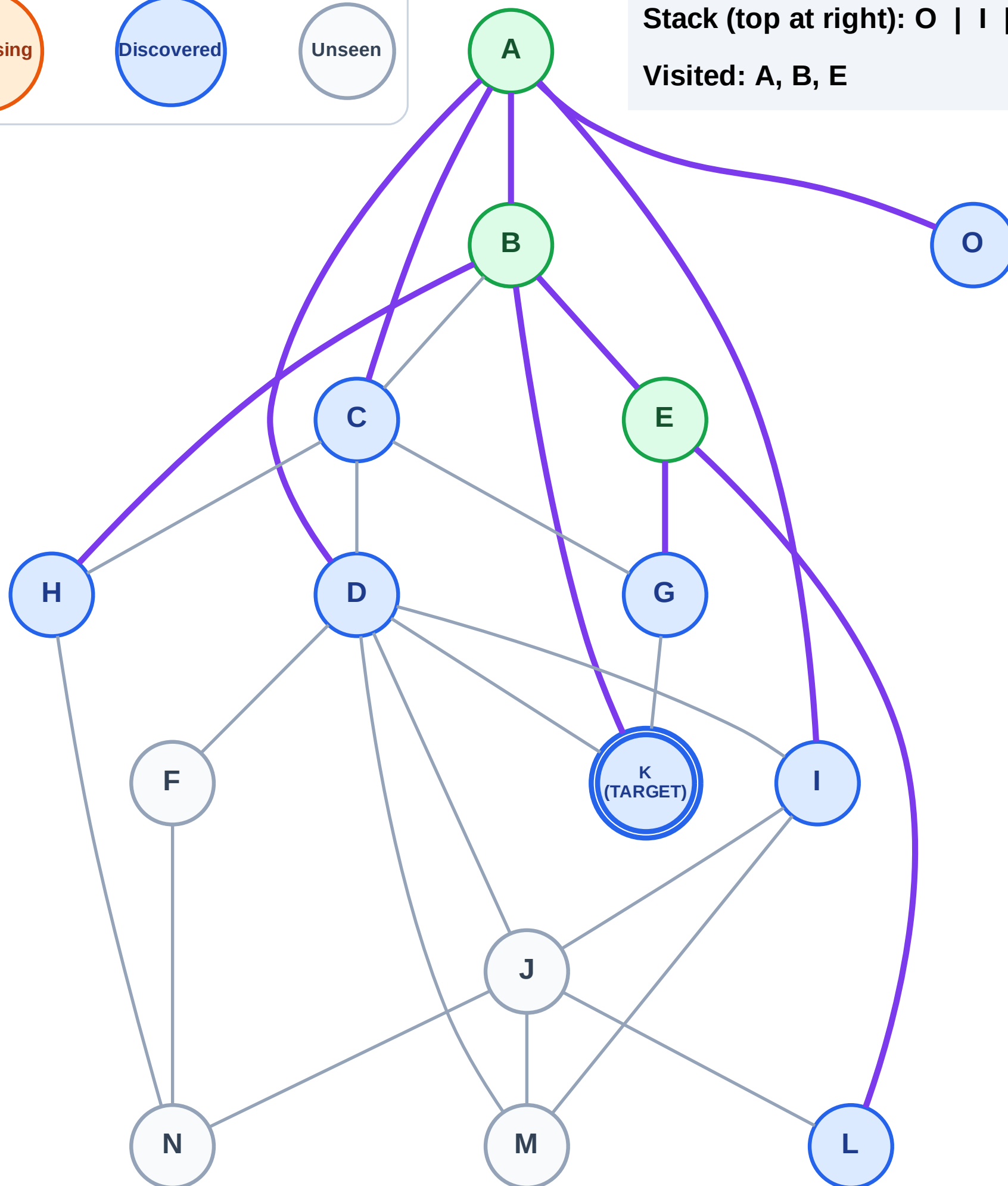
Visited: A, B



Step 7: Discover L, G from E

Stack (top at right): O | I | D | C | K | H | L | G

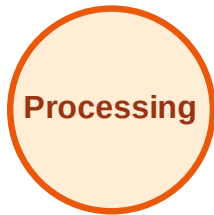
Visited: A, B, E



DFS Traversal

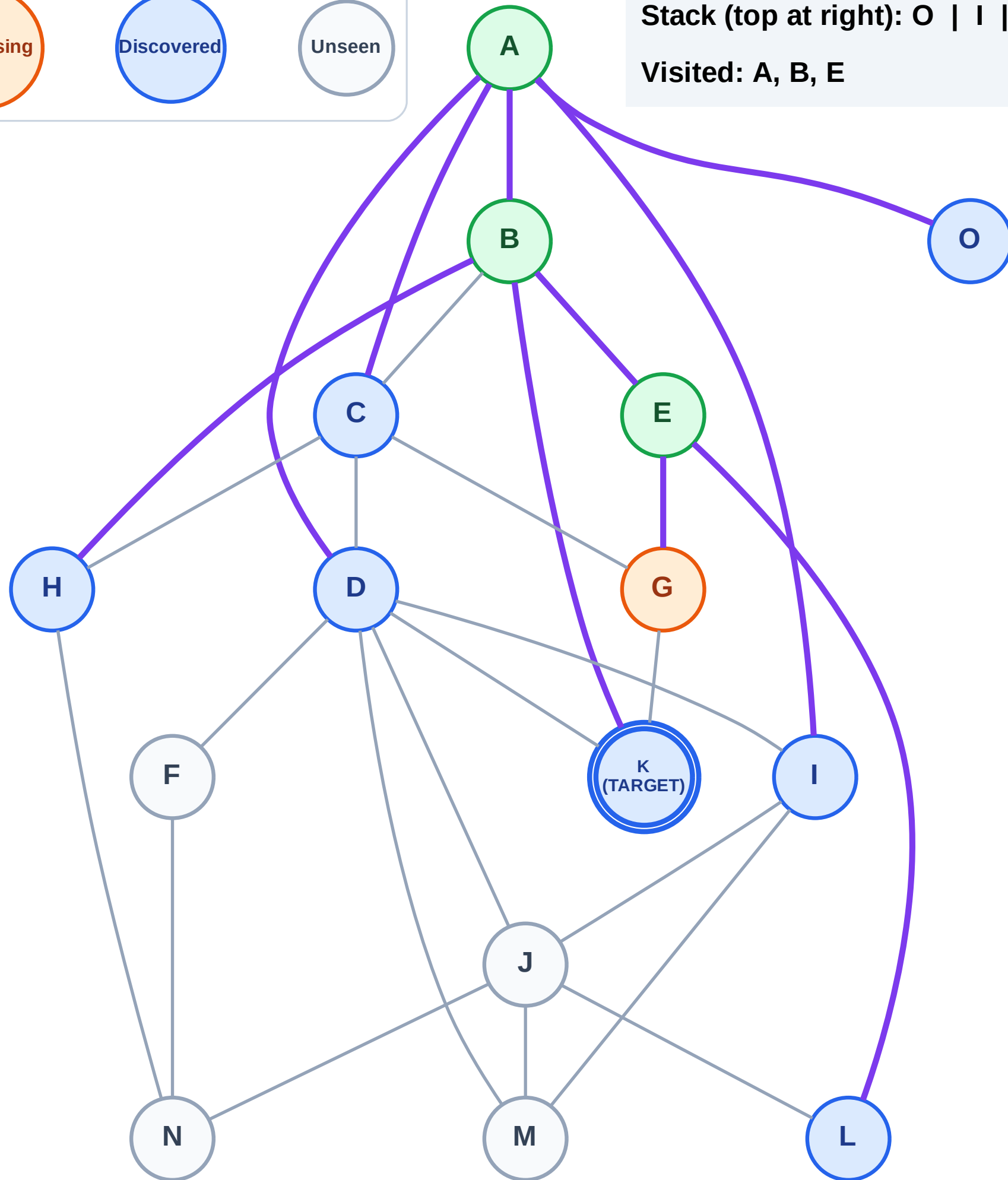
Step 8: Process node G

Legend



Stack (top at right): O | I | D | C | K | H | L

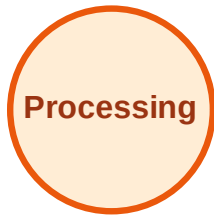
Visited: A, B, E



DFS Traversal

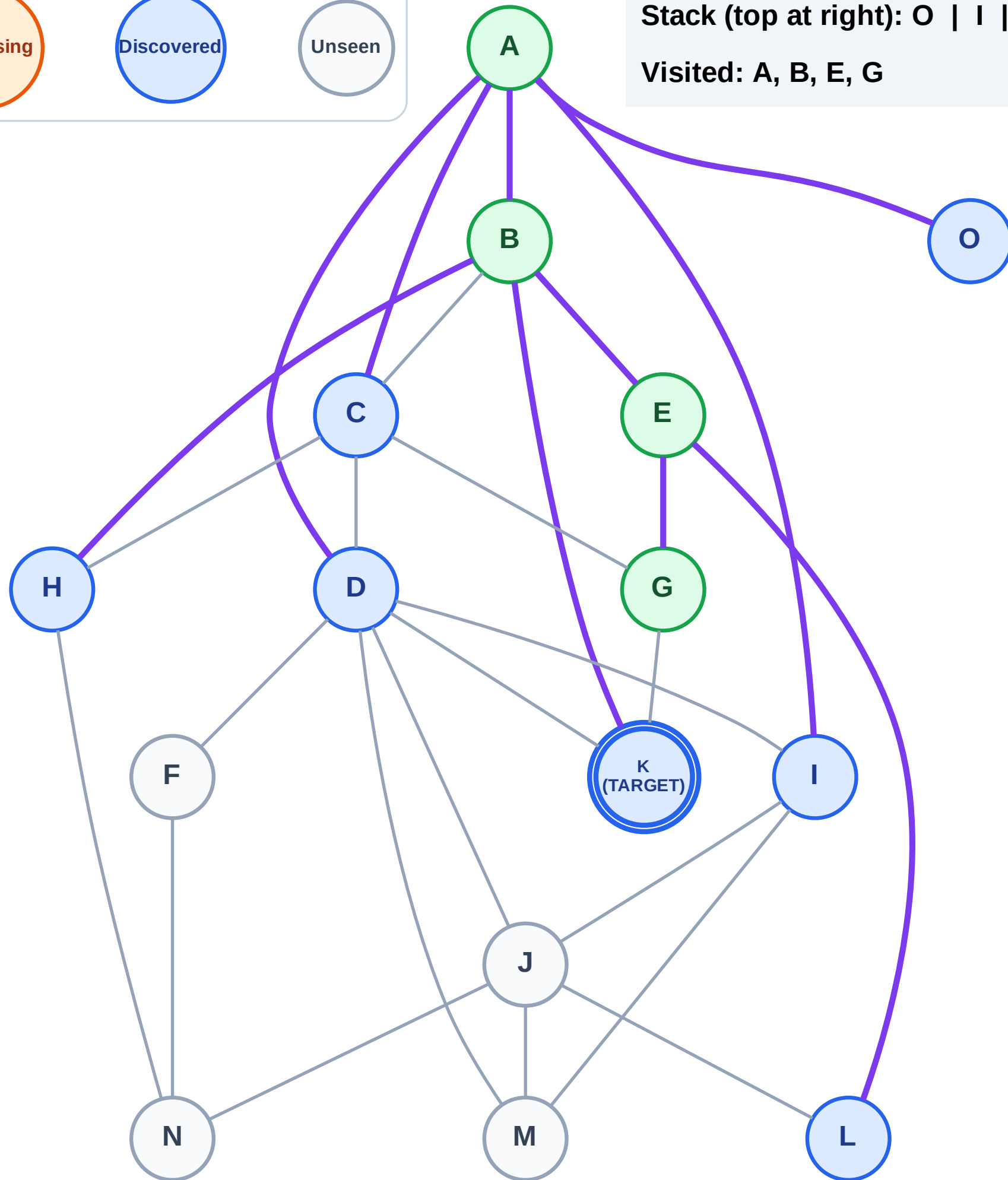
Step 9: No new neighbors from G

Legend



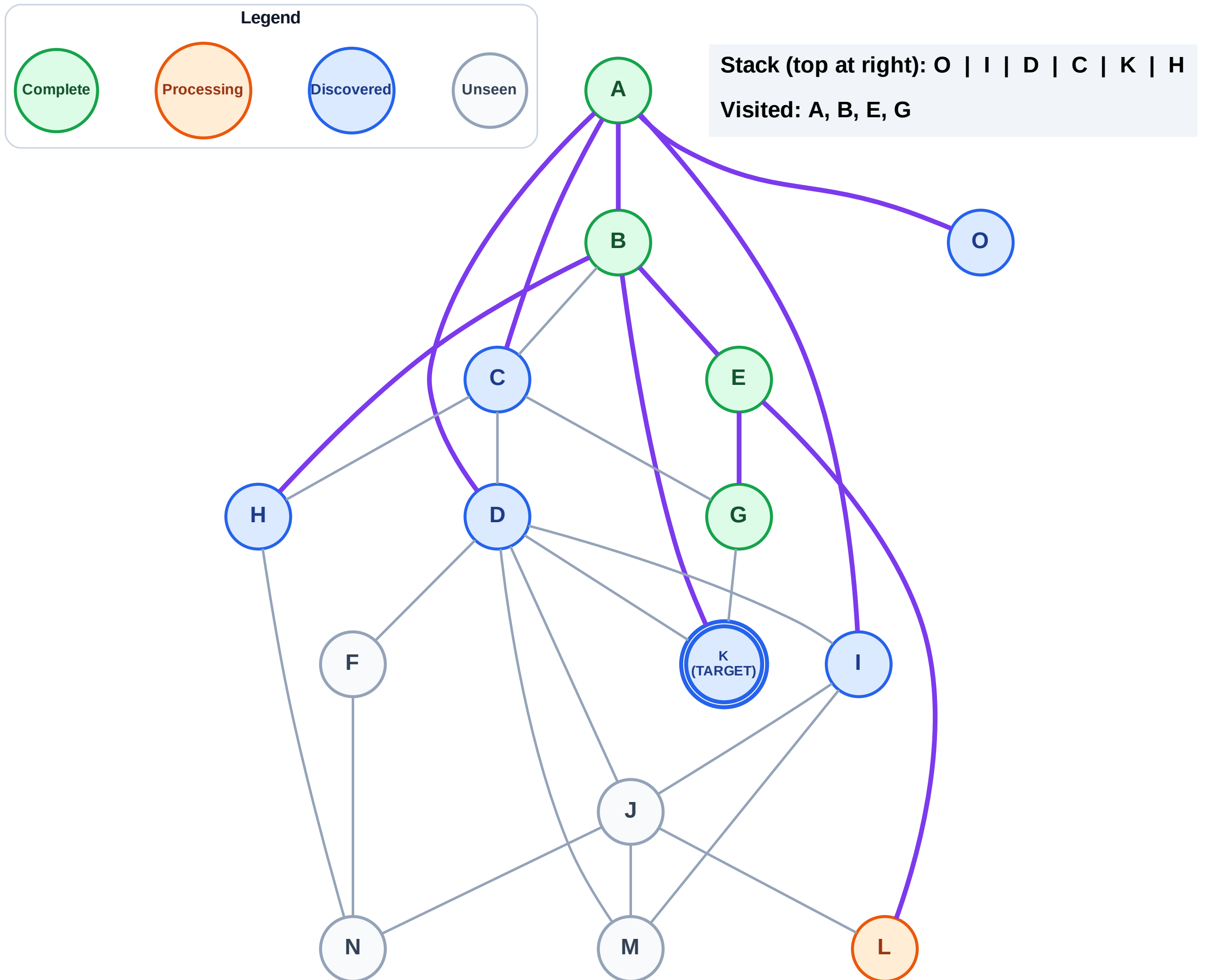
Stack (top at right): O | I | D | C | K | H | L

Visited: A, B, E, G



Step 10: Process node L

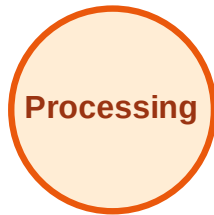
Step 10: Process node L



DFS Traversal

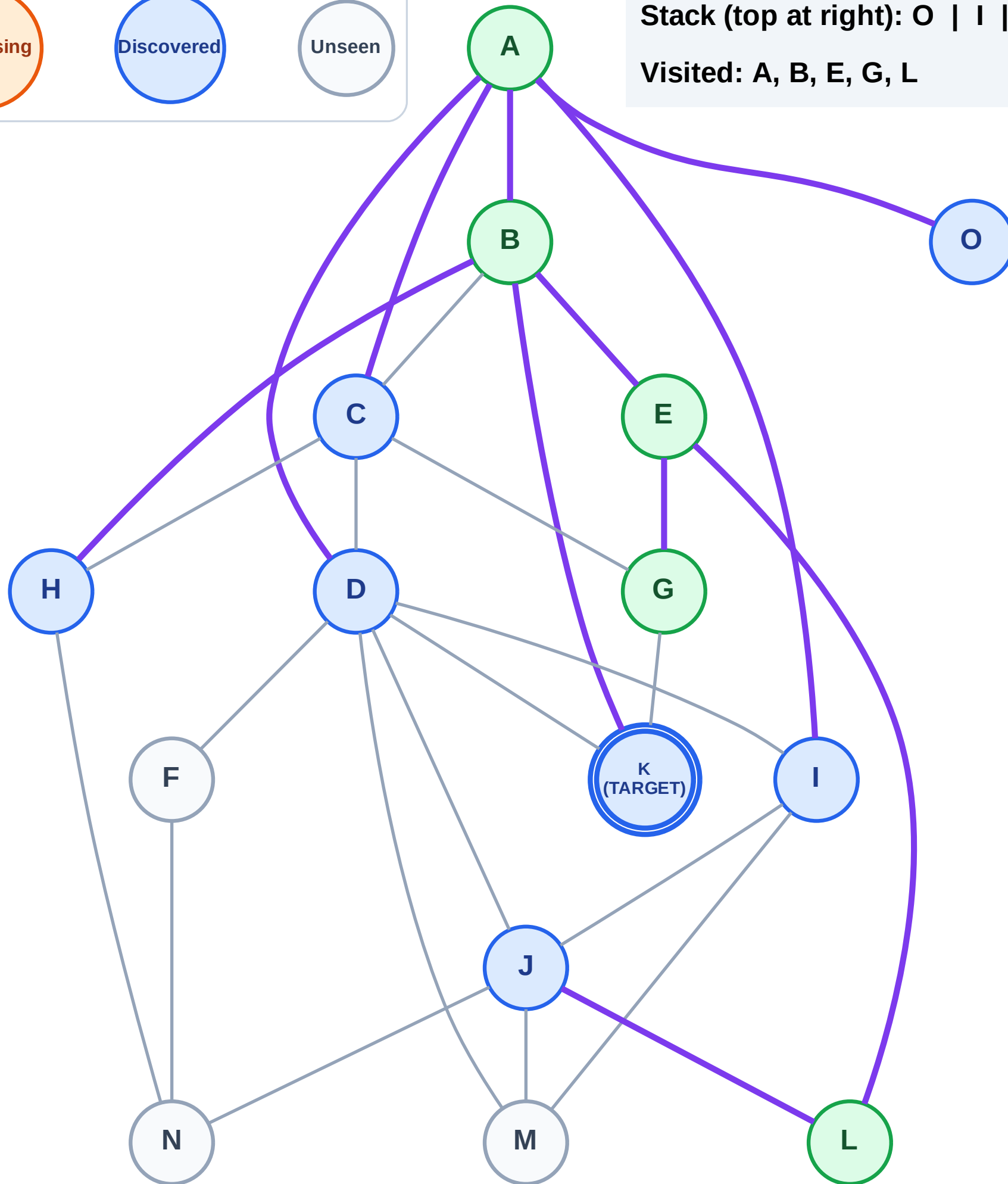
Step 11: Discover J from L

Legend



Stack (top at right): O | I | D | C | K | H | J

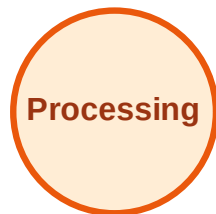
Visited: A, B, E, G, L



DFS Traversal

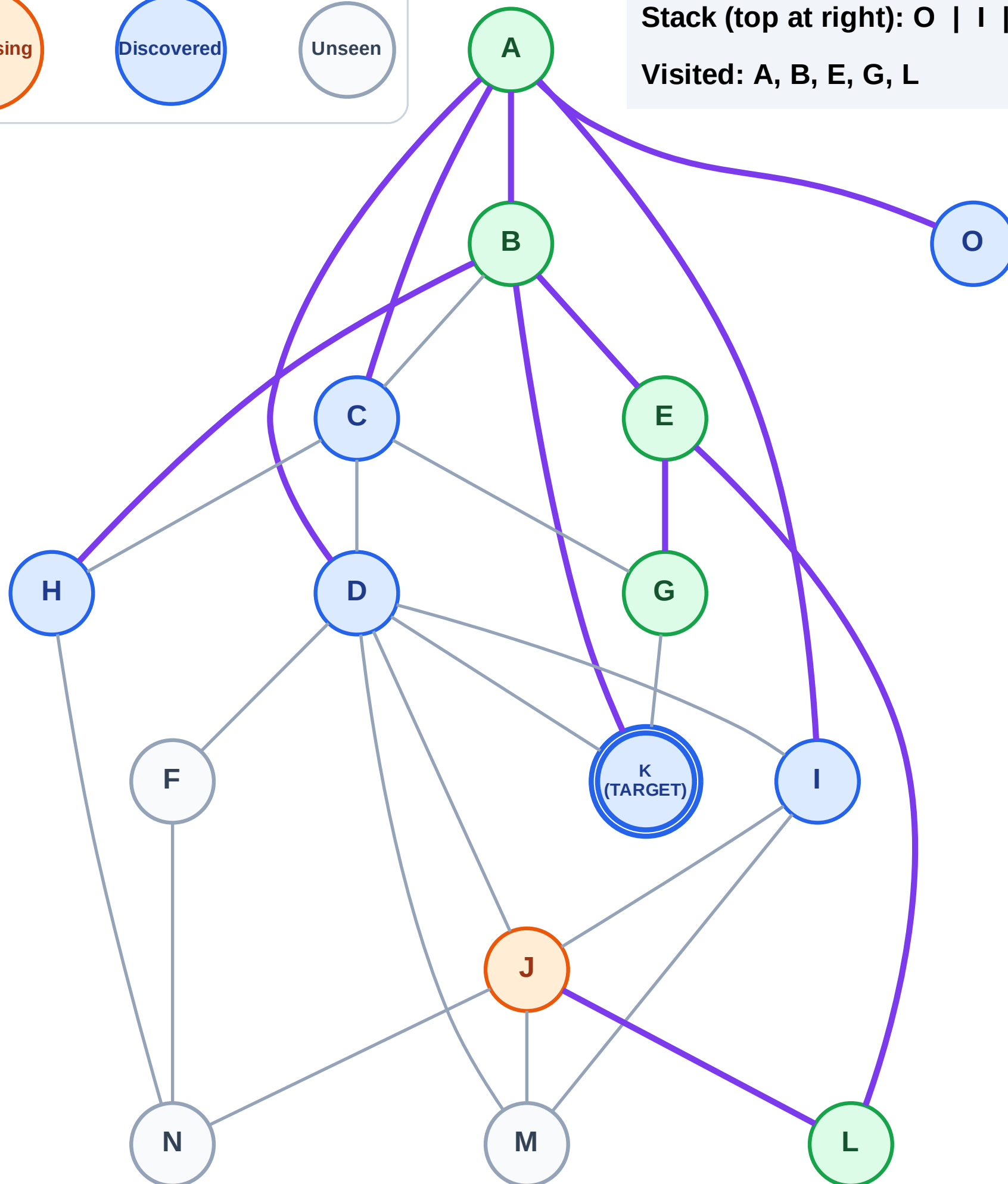
Step 12: Process node J

Legend



Stack (top at right): O | I | D | C | K | H

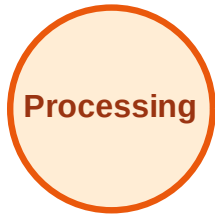
Visited: A, B, E, G, L



DFS Traversal

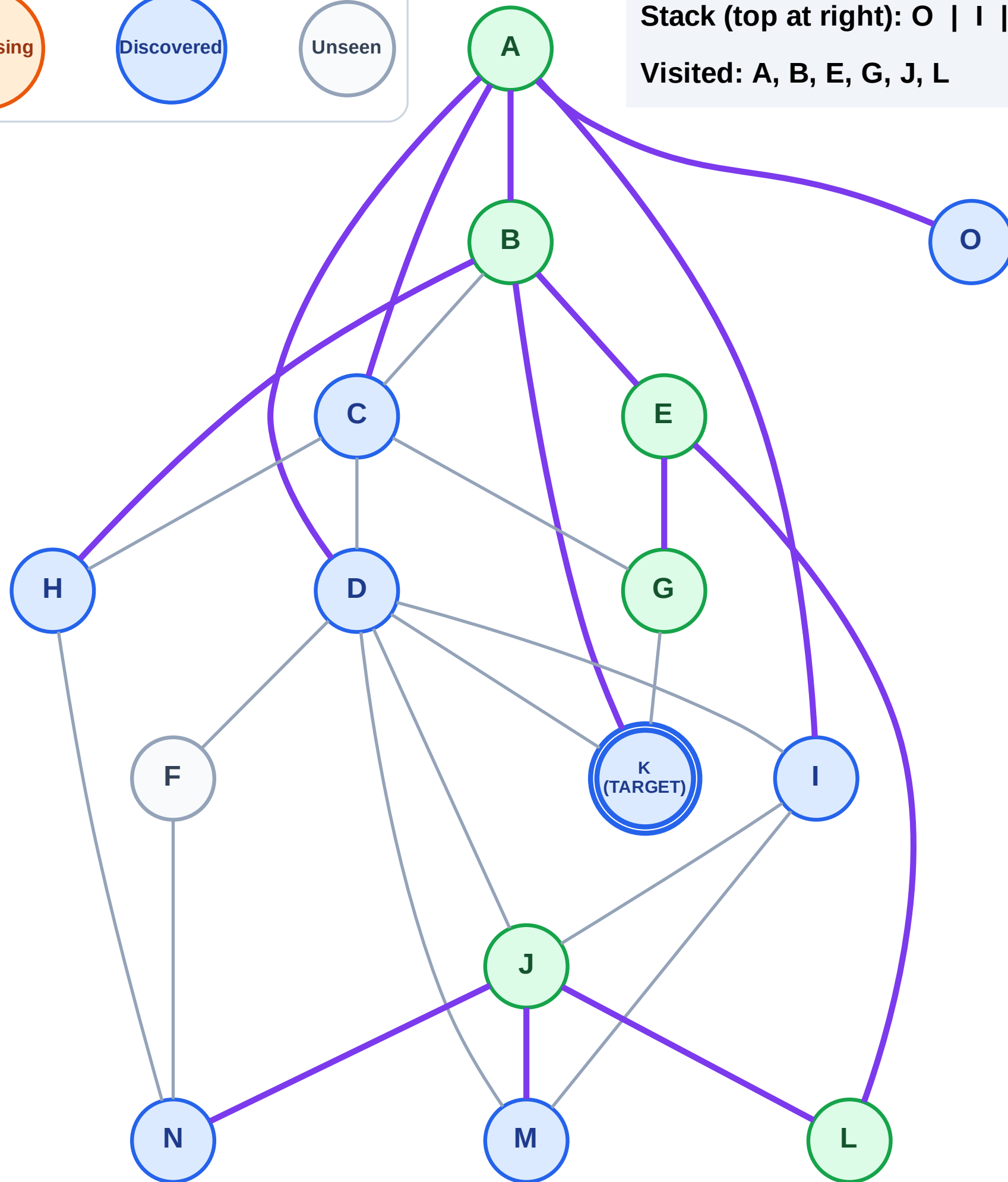
Step 13: Discover N, M from J

Legend



Stack (top at right): O | I | D | C | K | H | N | M

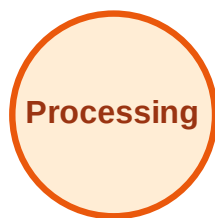
Visited: A, B, E, G, J, L



DFS Traversal

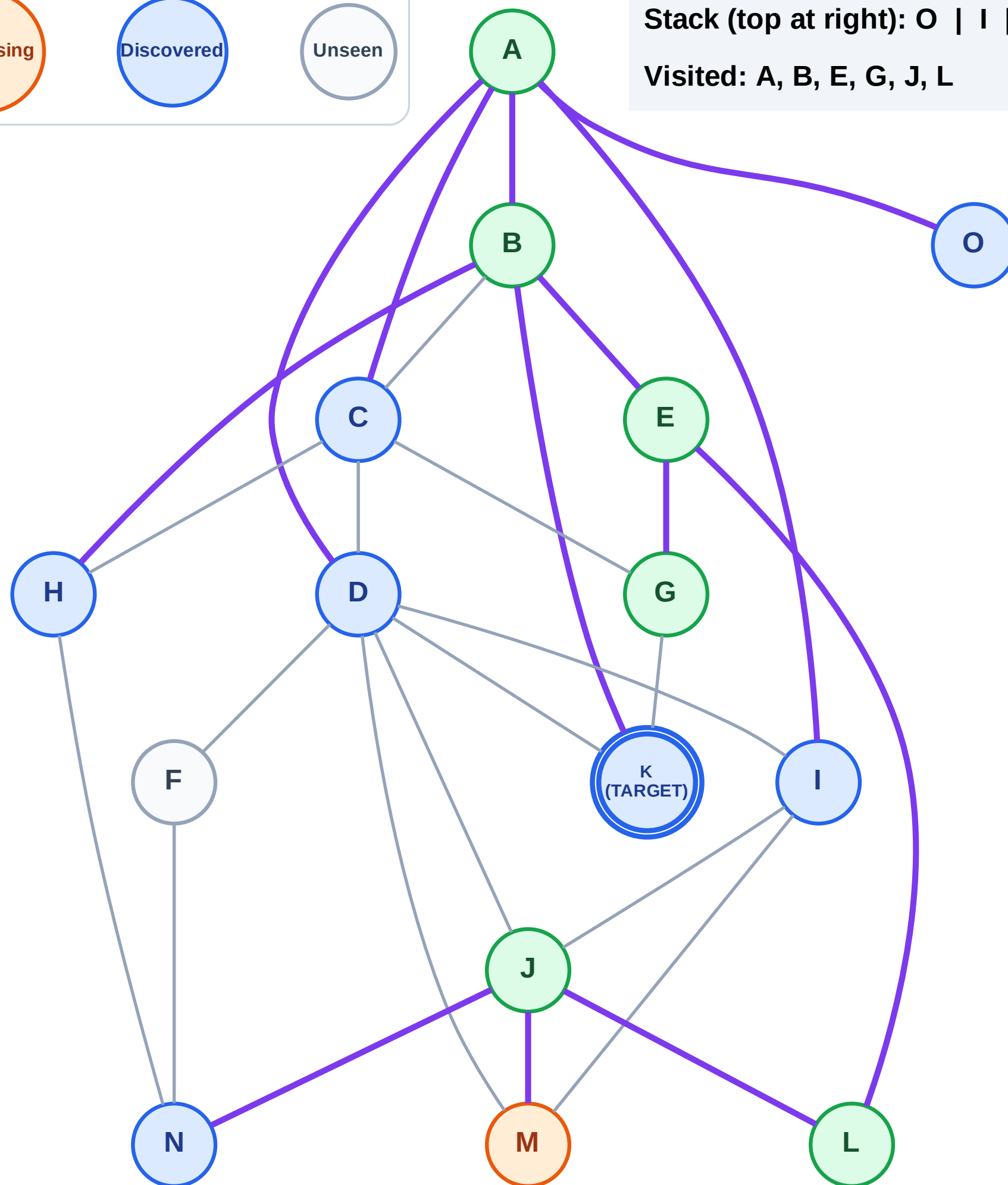
Step 14: Process node M

Legend



Stack (top at right): O | I | D | C | K | H | N

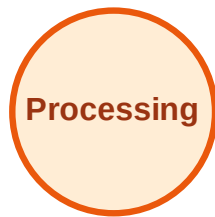
Visited: A, B, E, G, J, L



DFS Traversal

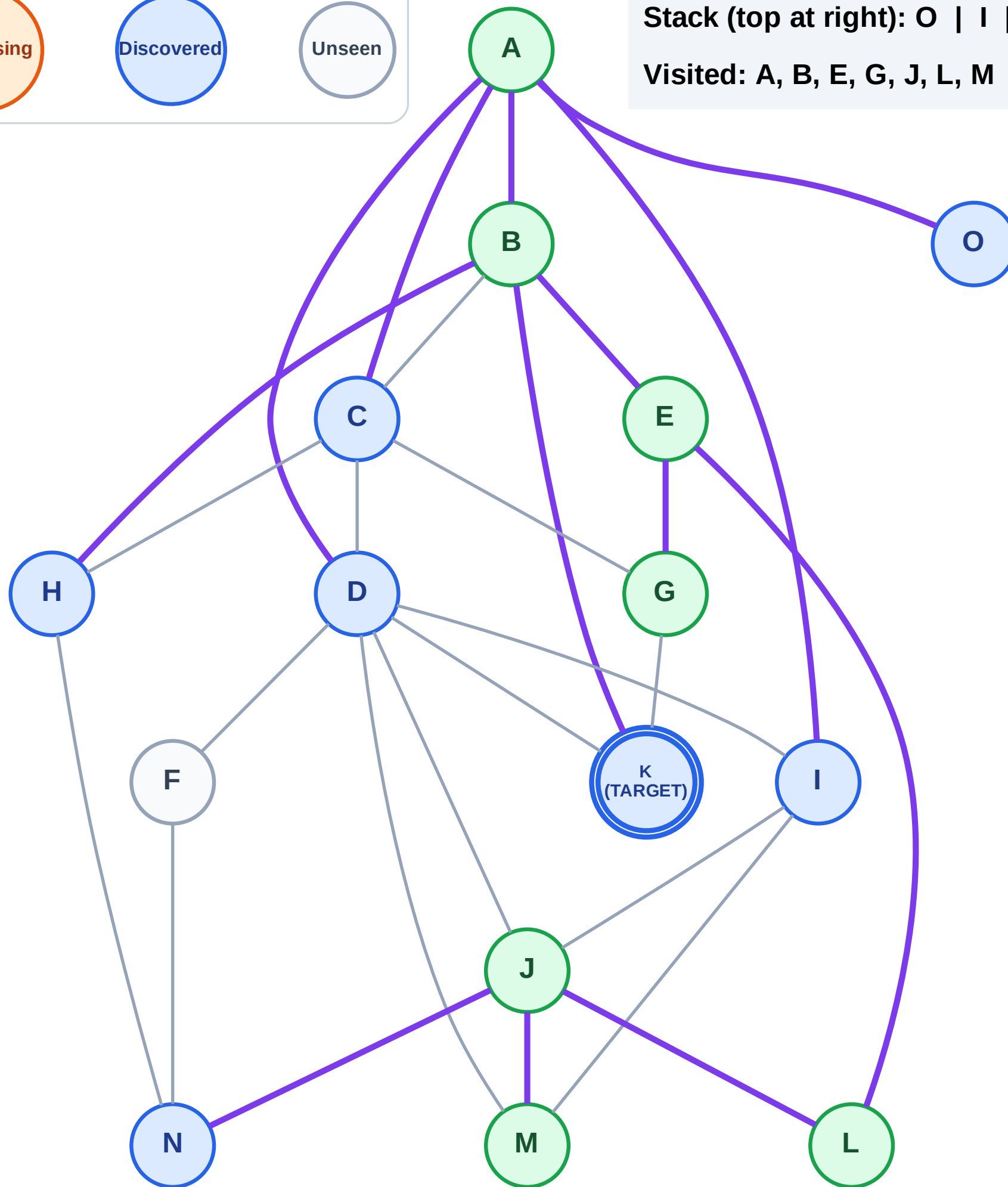
Step 15: No new neighbors from M

Legend



Stack (top at right): O | I | D | C | K | H | N

Visited: A, B, E, G, J, L, M



DFS Traversal

Step 16: Process node N

Legend

Complete

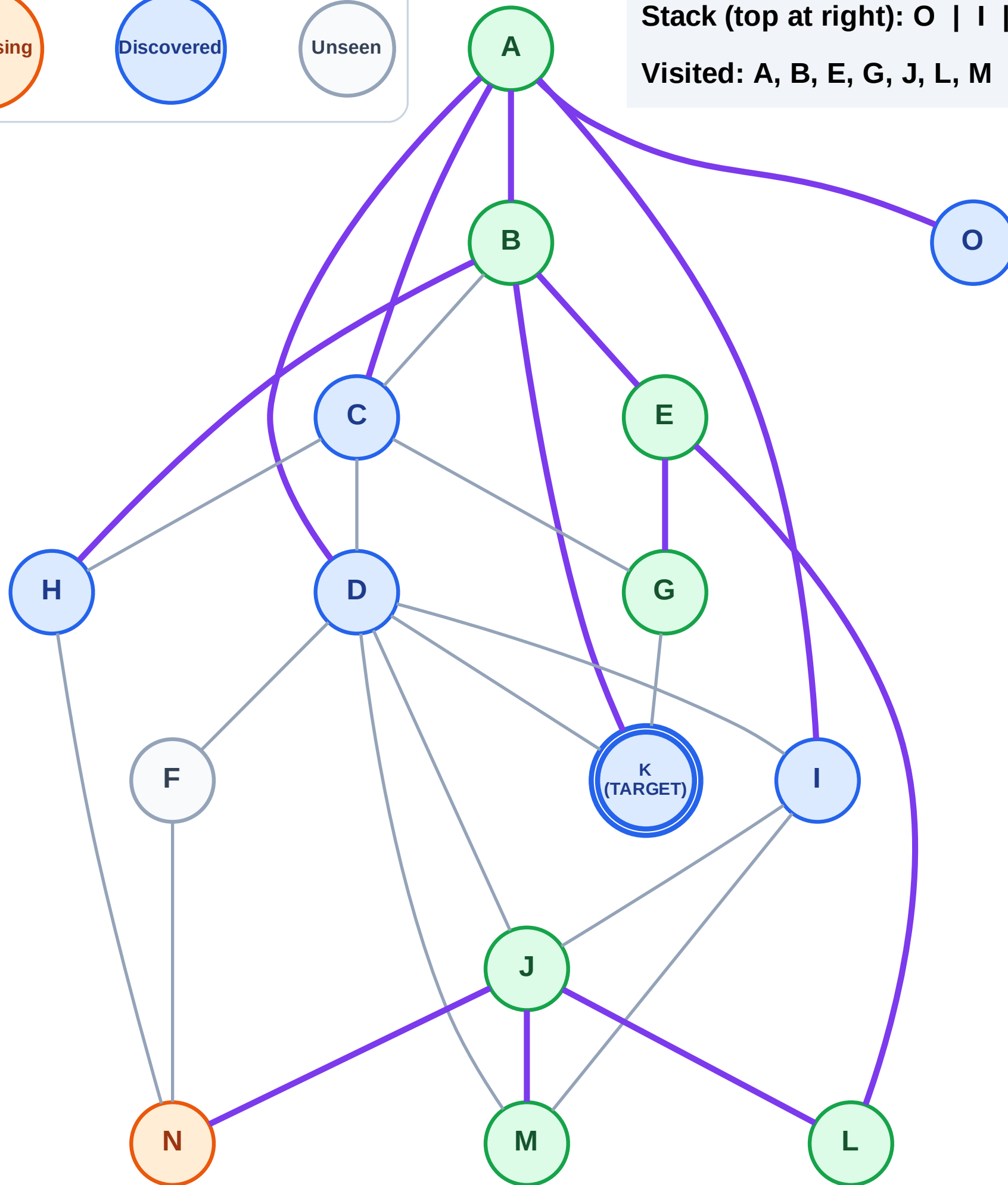
Processing

Discovered

Unseen

Stack (top at right): O | I | D | C | K | H

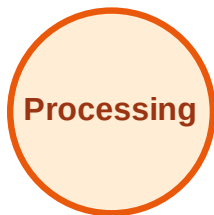
Visited: A, B, E, G, J, L, M



DFS Traversal

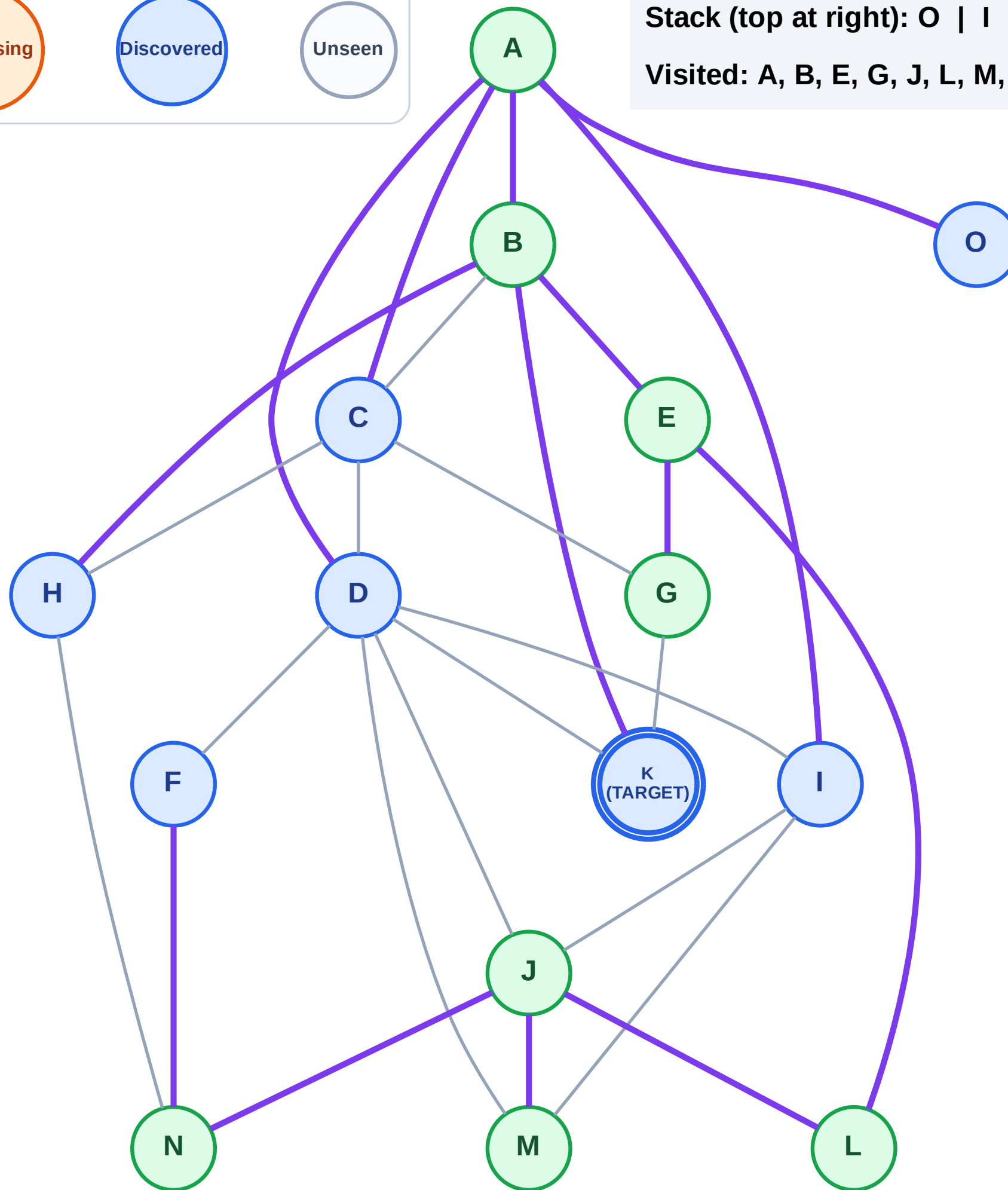
Step 17: Discover F from N

Legend



Stack (top at right): O | I | D | C | K | H | F

Visited: A, B, E, G, J, L, M, N



DFS Traversal

Step 18: Process node F

Legend

Complete

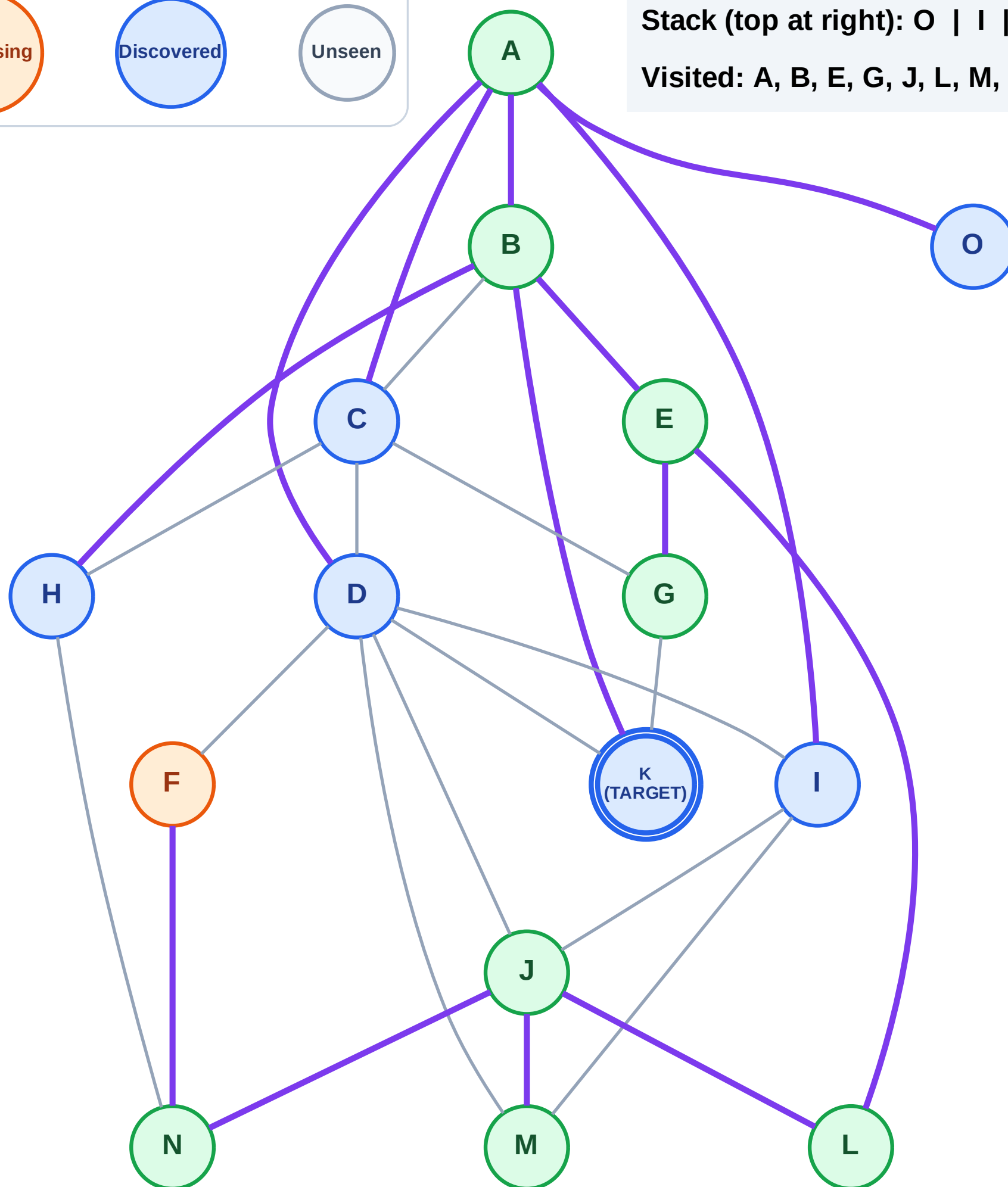
Processing

Discovered

Unseen

Stack (top at right): O | I | D | C | K | H

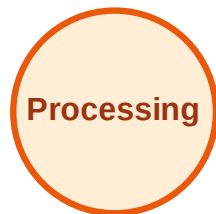
Visited: A, B, E, G, J, L, M, N



DFS Traversal

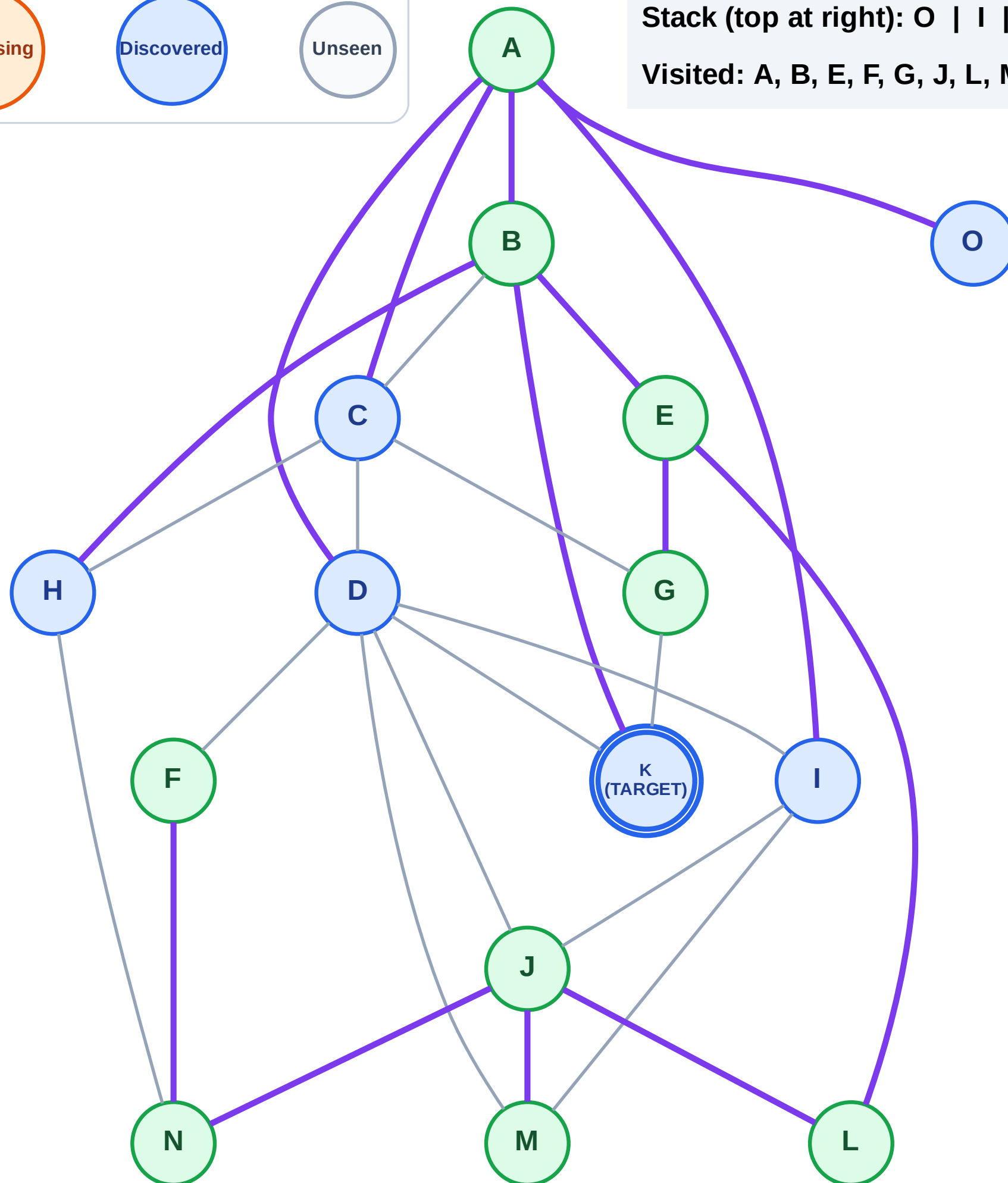
Step 19: No new neighbors from F

Legend



Stack (top at right): O | I | D | C | K | H

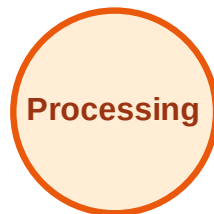
Visited: A, B, E, F, G, J, L, M, N



DFS Traversal

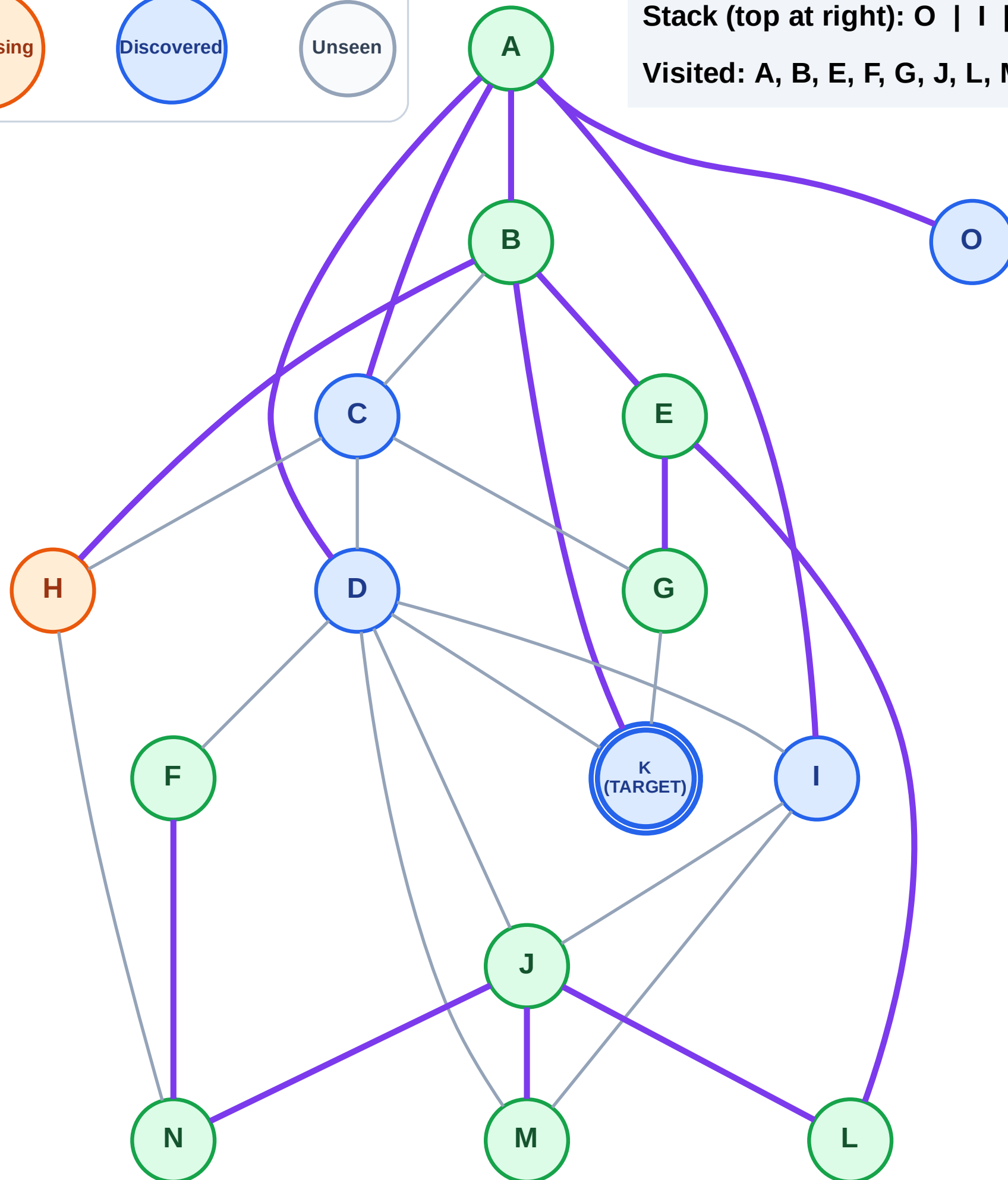
Step 20: Process node H

Legend



Stack (top at right): O | I | D | C | K

Visited: A, B, E, F, G, J, L, M, N



DFS Traversal

Step 21: No new neighbors from H

Legend

Complete

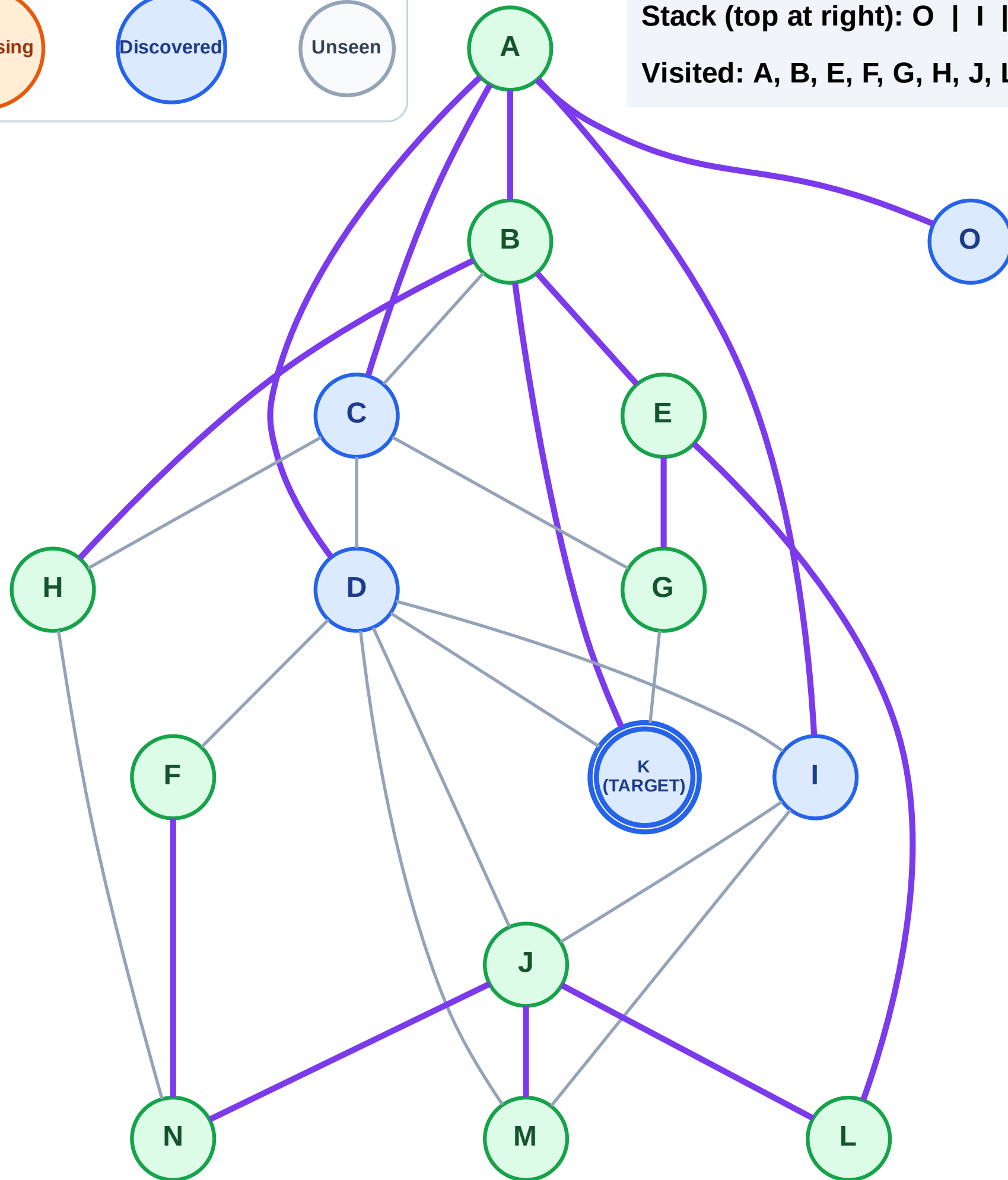
Processing

Discovered

Unseen

Stack (top at right): O | I | D | C | K

Visited: A, B, E, F, G, H, J, L, M, N



DFS Traversal

Step 22: Process node K

Legend

Complete

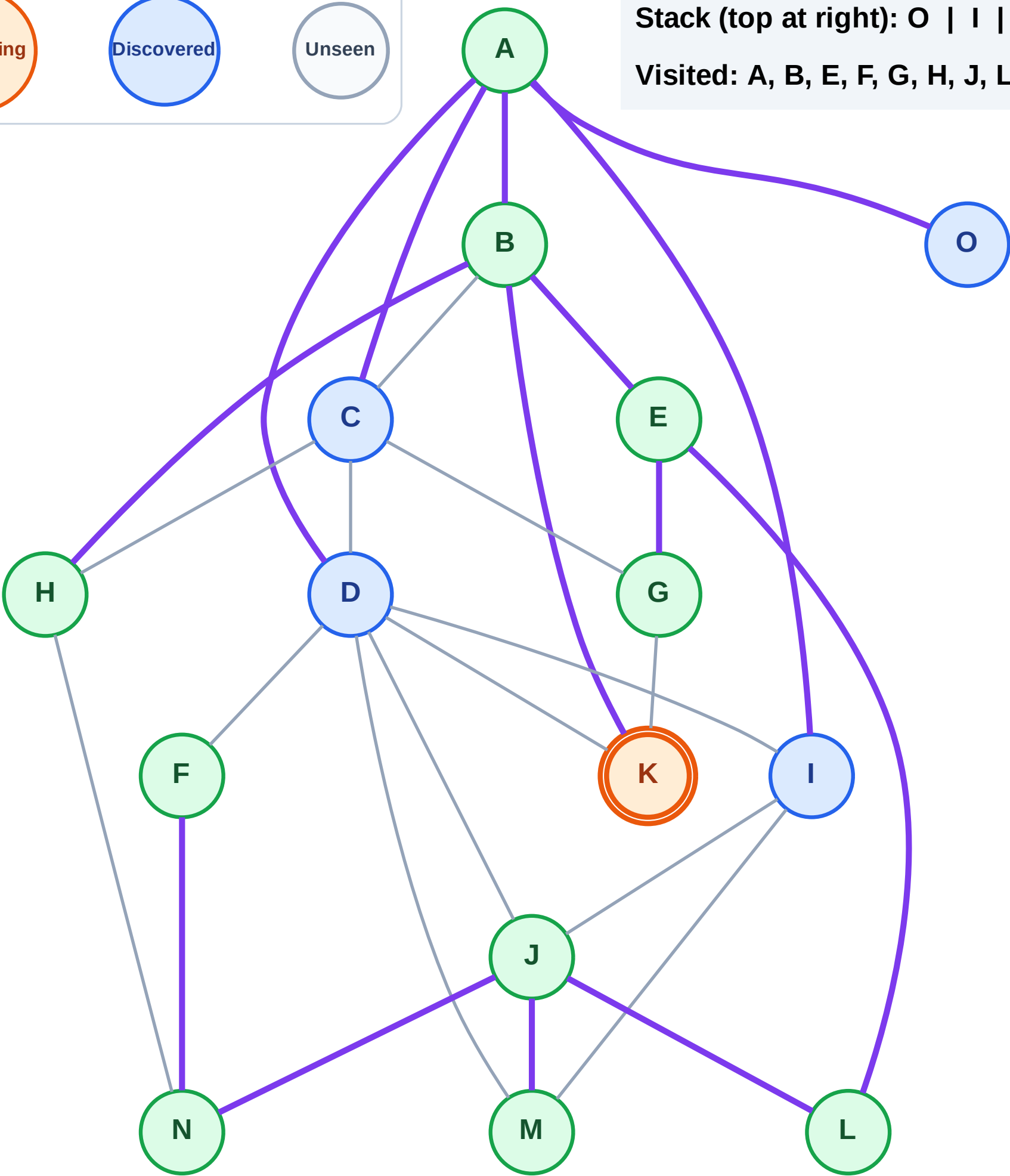
Processing

Discovered

Unseen

Stack (top at right): O | I | D | C

Visited: A, B, E, F, G, H, J, L, M, N



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | I | D | C
Visited: A, B, E, F, G, H, J, K, L, M, N

